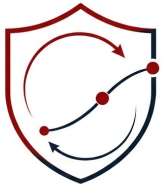
Principles of Robot Autonomy I

Planning and Control: A* path planning, RRT and sampling based planning

Stanford University



SIA Lab
Safe and Intelligent Autonomy Lab

IPRL

Principles of Robot Autonomy I

Announcements:

- Lab section 1 this week
- HW 1 due next Thurs, Oct 8
- Slides uploaded to canvas before lecture
- Total enrolled: 160, total waitlist: 70

Lecture 4:

- Dijkstras and A* graph planning
- PRM
- RRT, RRT* and re-wiring

Stanford University

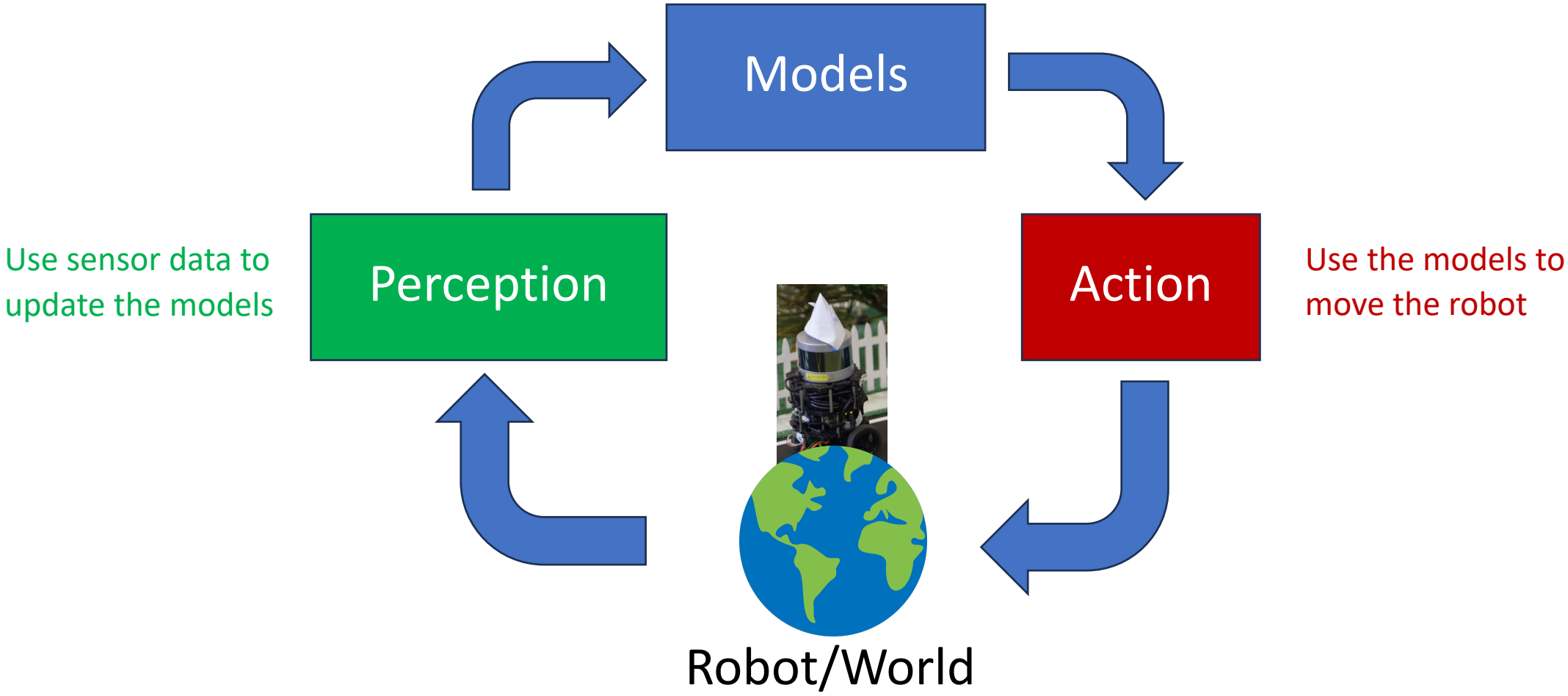
SIA Lab
Safe and Intelligent Autonomy Lab

MSL

IPRL

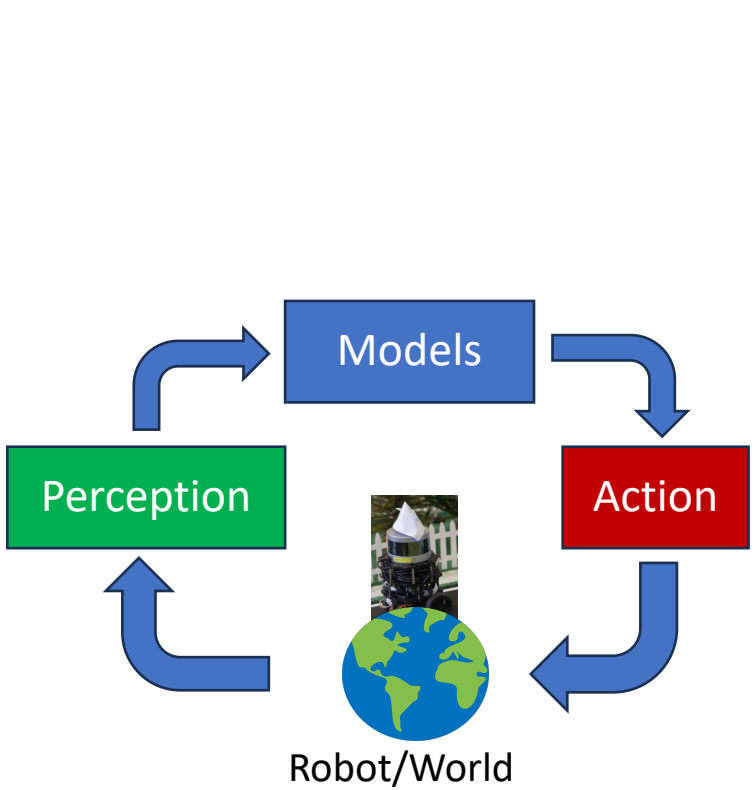
ASL
Autonomous Systems Lab

Full Stack Autonomy: the Perception-Action Loop



Action: Planning and Control Stack

Use the models to move the robot



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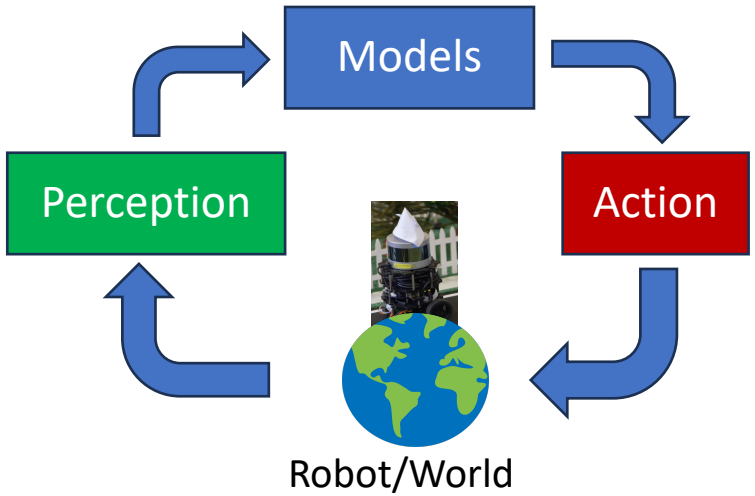
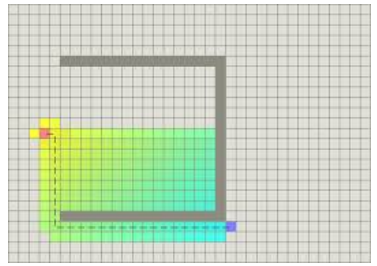
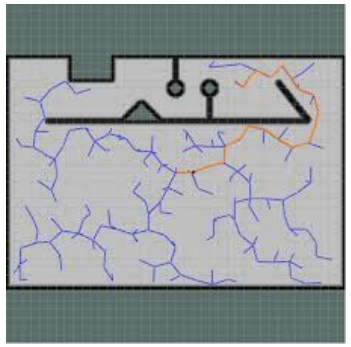


Path planning:

- A* planning on a graph
- RRT, RRT*





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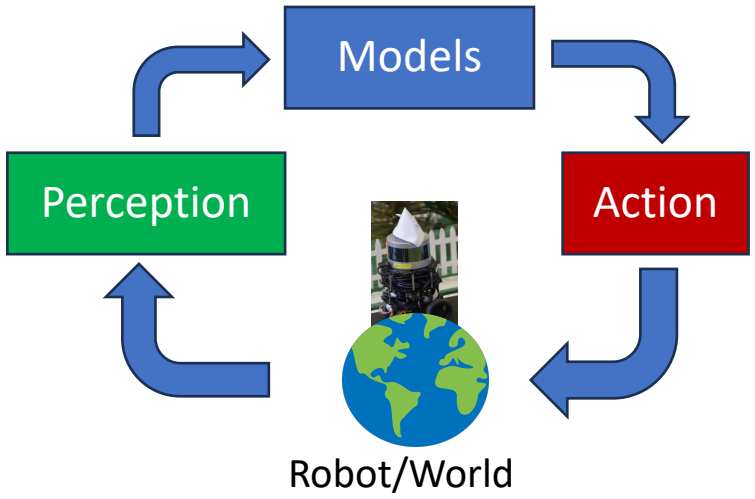

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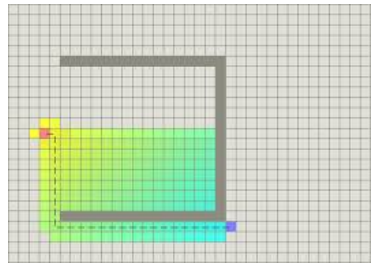
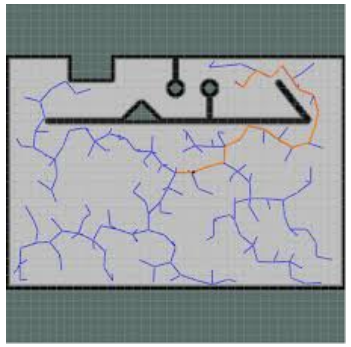
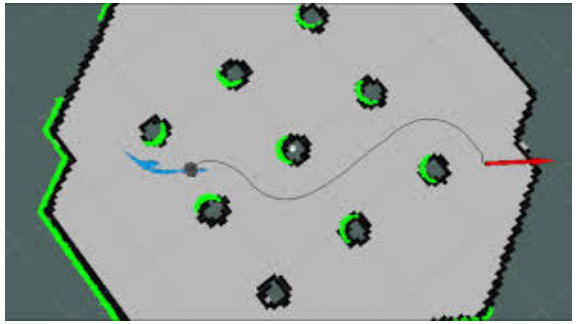




Trajectory optimization:

- Optimal control
- Numerical solvers



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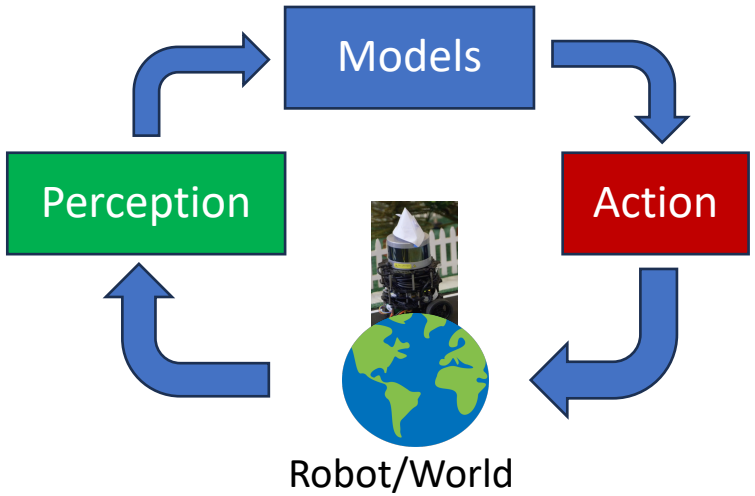

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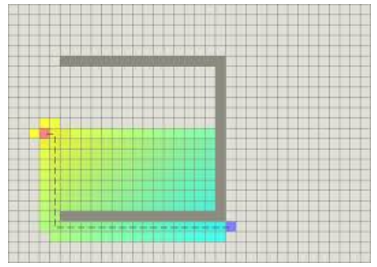

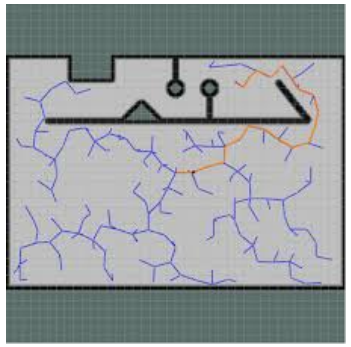

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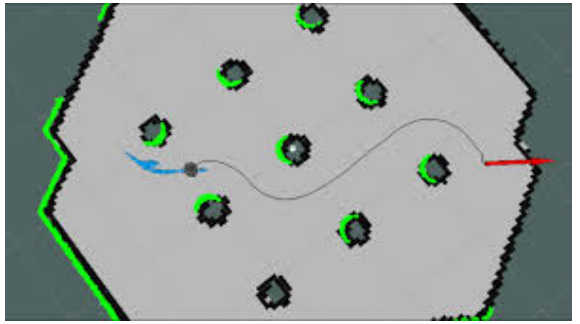

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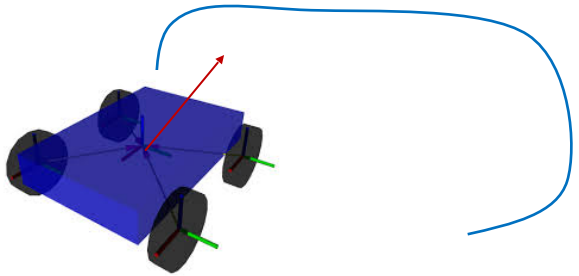

Path Planning

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Map representation

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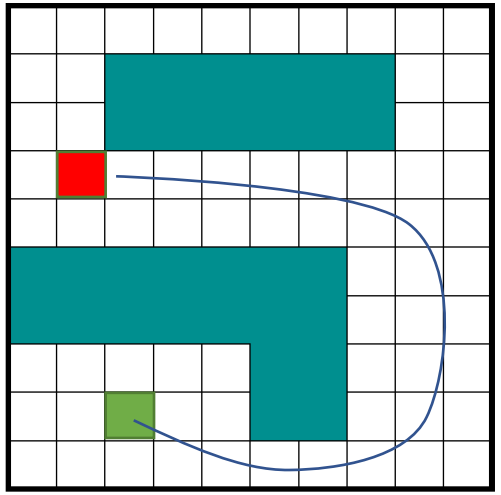
Graph search planning, A*



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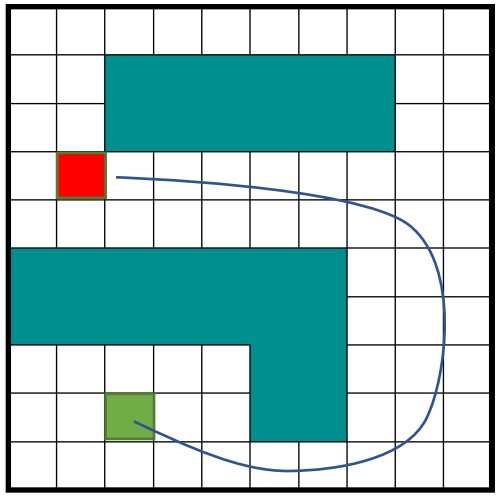
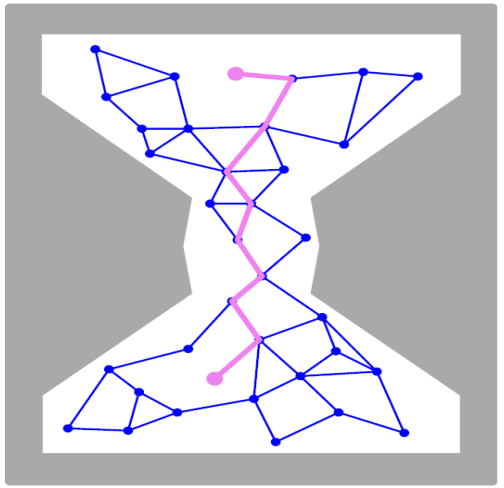


Planning on random graph, RRT*, PRM



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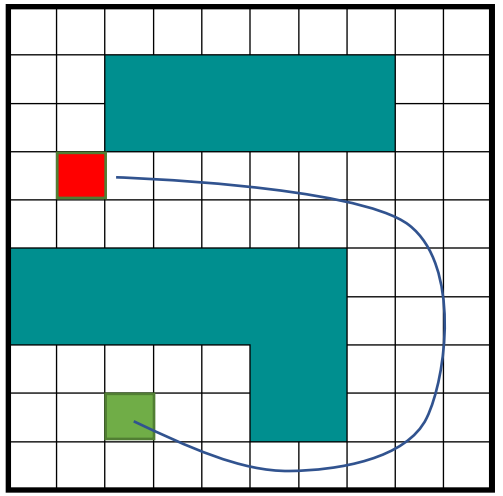

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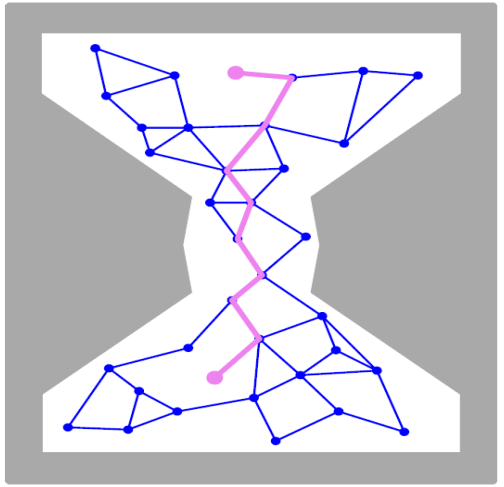
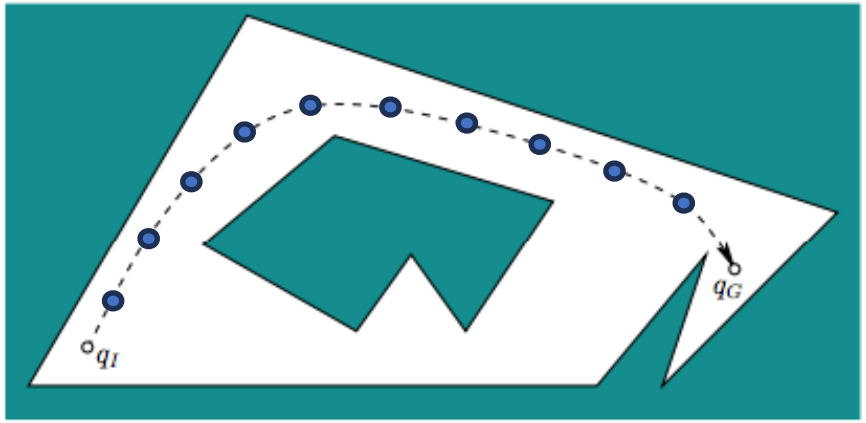


Goal: Find a path so the robot doesn't hit things!

Trajectory Optimization



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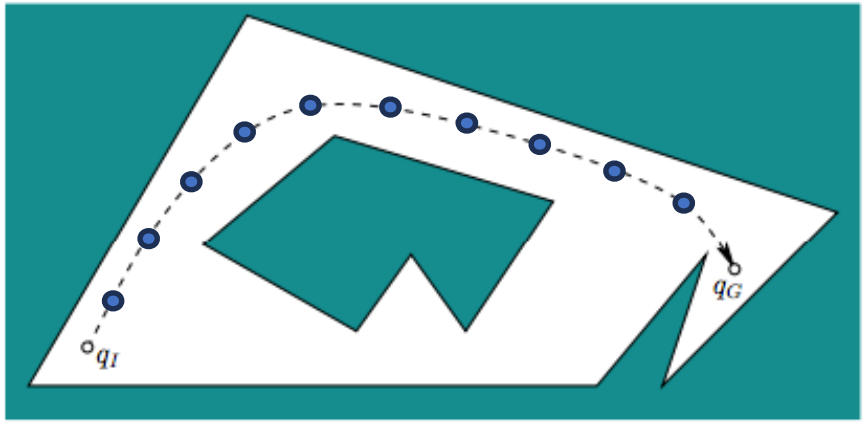

Trajectory optimization problem:

$$\min_{u_{0:T}} \sum_{t=0}^{T} c_t(x_t, u_t)$$

$$\text{s.t.} \quad x_{t+1} = f(x_t, u_t)$$

$$x_t \in \mathbb{X}_{\text{Free}}$$

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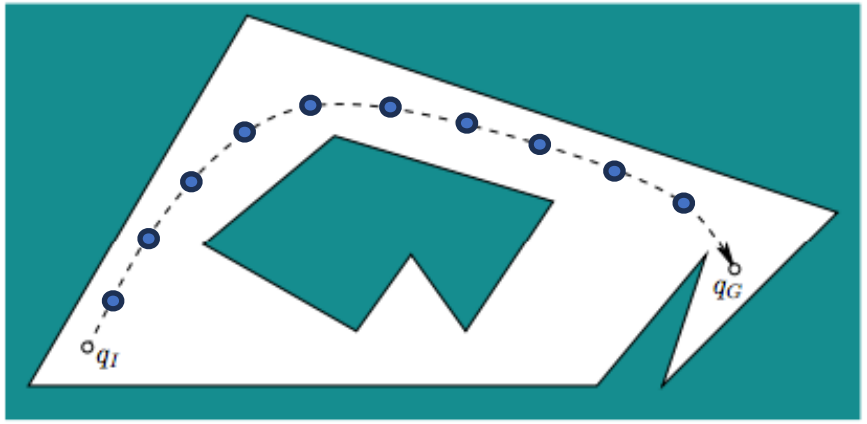

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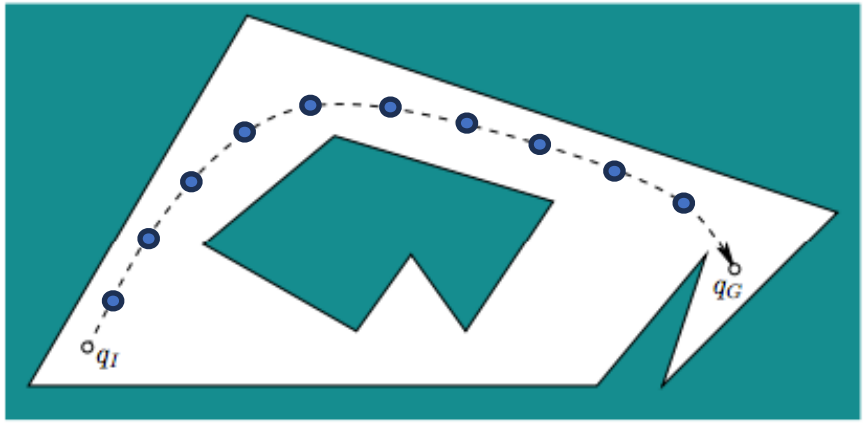

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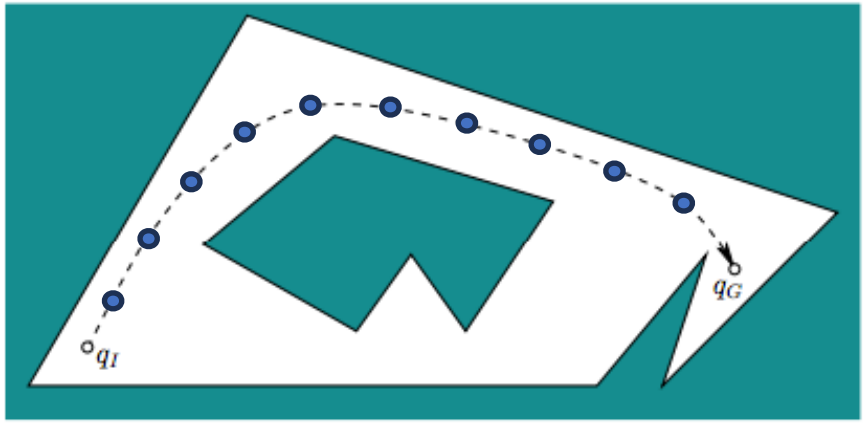

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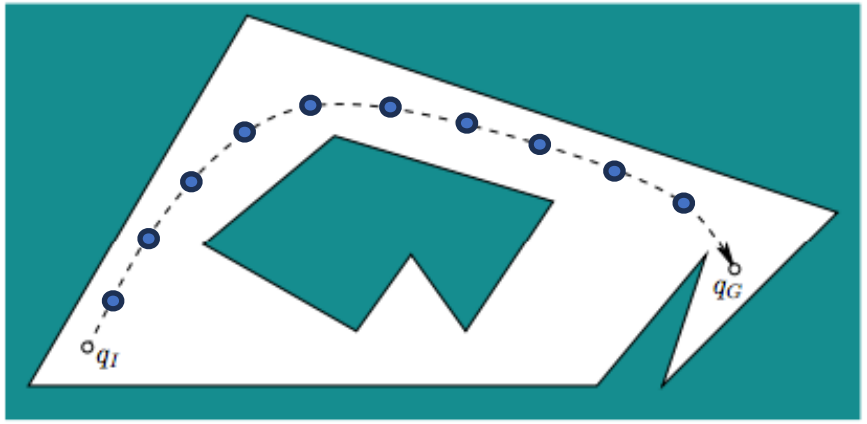

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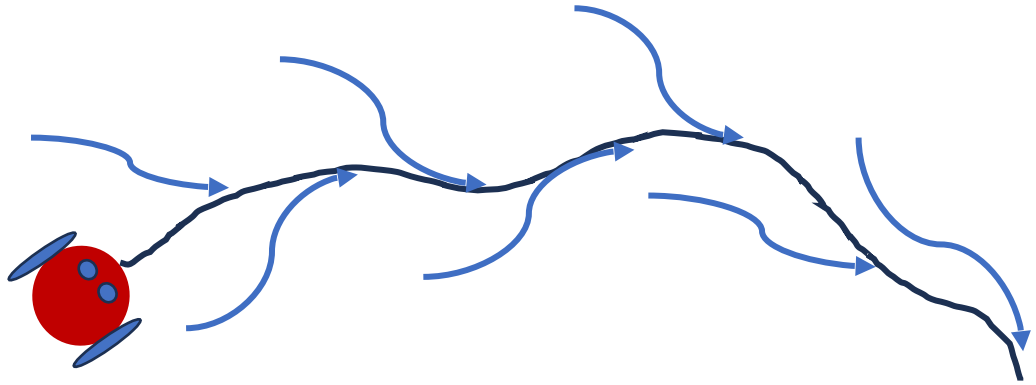
Goal: Find a dynamically feasible trajectory (close to the path) so the robot doesn't hit things!

Trajectory Following



Motion model:

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Feedback control law:

$$u_t = h(x_t, x_t^*)$$

planned traj waypoints

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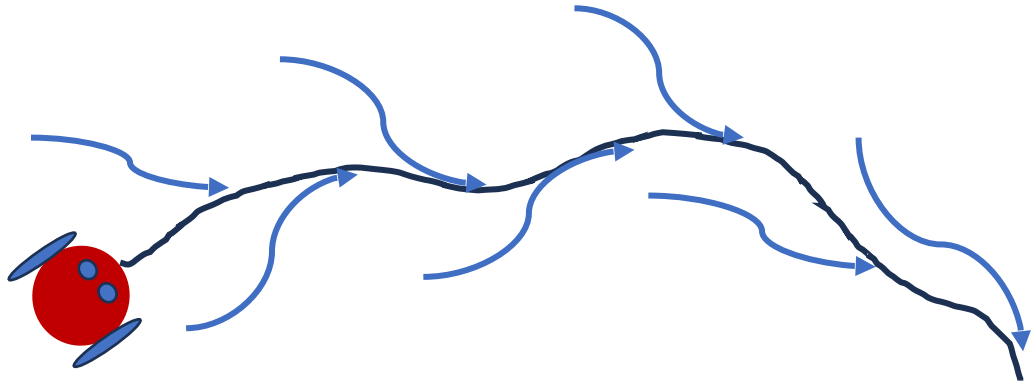

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Goal: Use closed loop feedback control to track desired trajectory, rejecting disturbances and being robust to model errors.

Path Planning vs Trajectory Planning

A* path plan:

(i) Global search
(ii) Collision free
(iii) Not dynamically feasible

Traj Opt

(i) Local search
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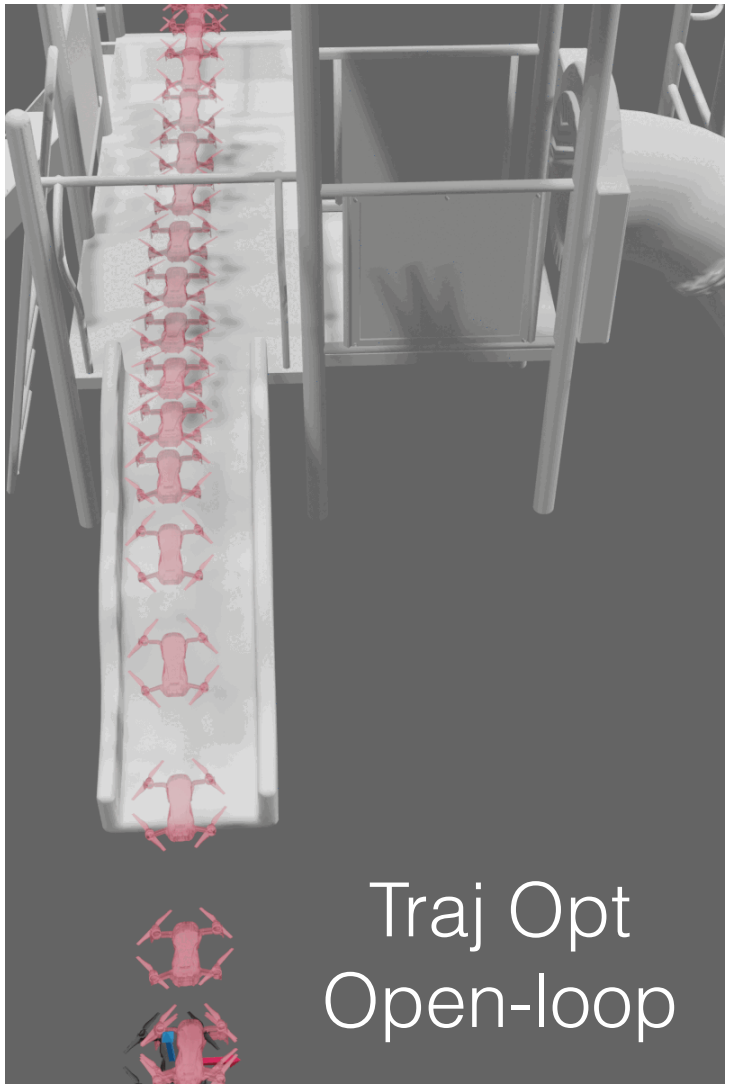

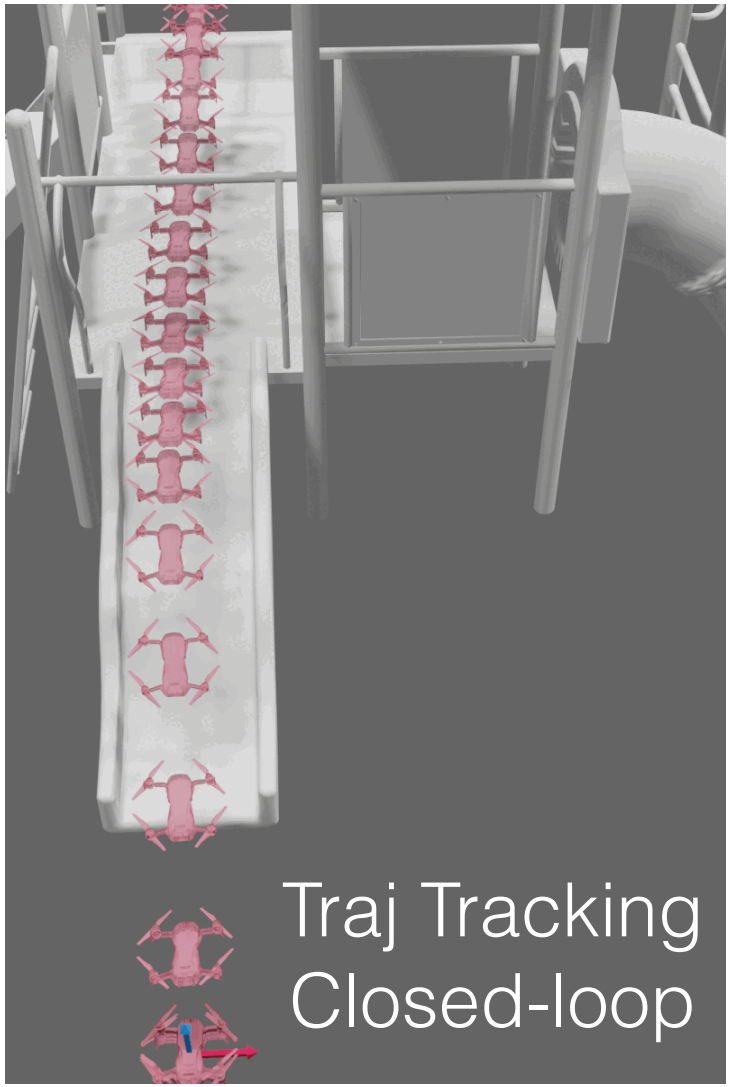

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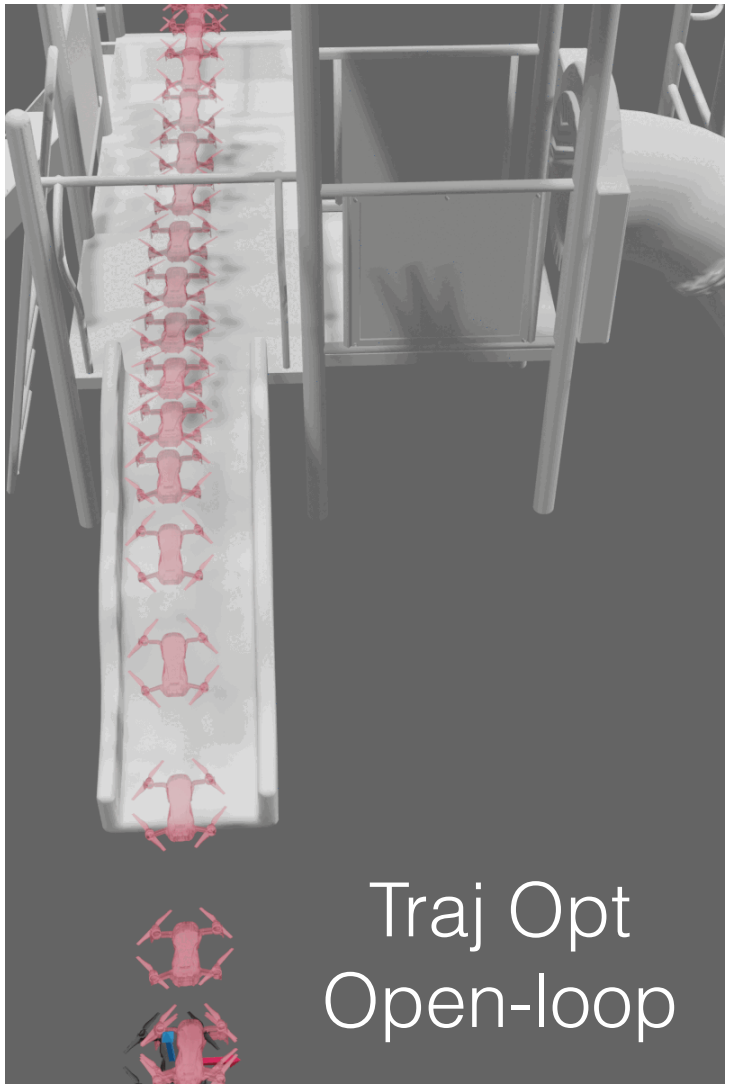

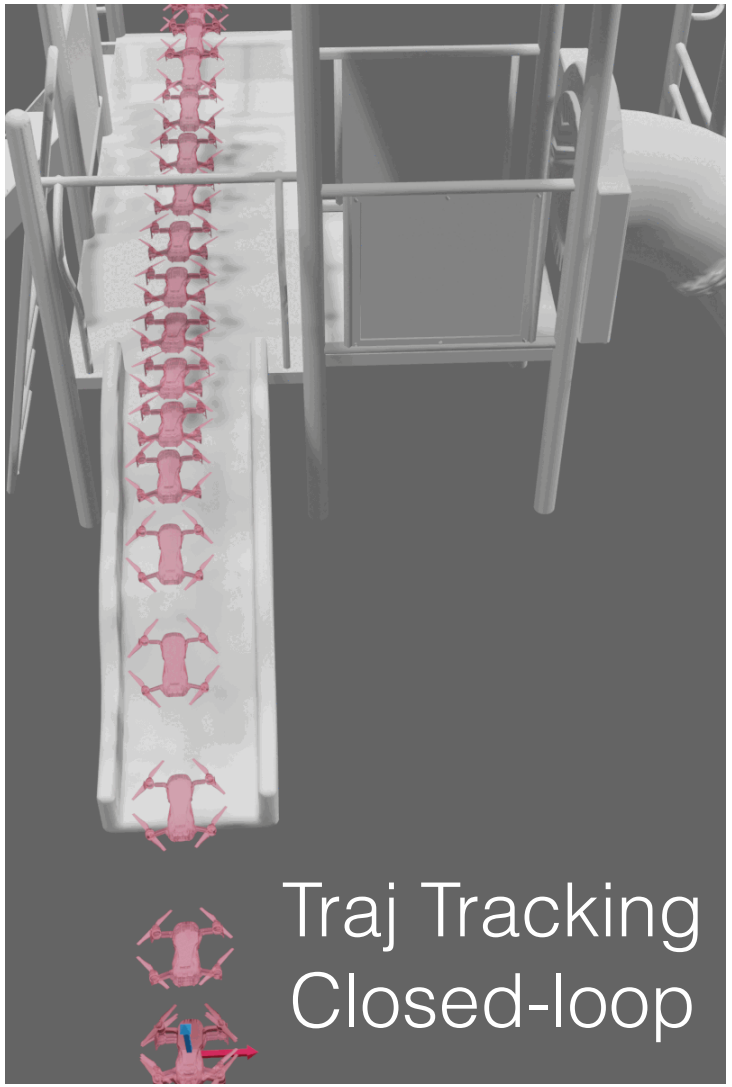

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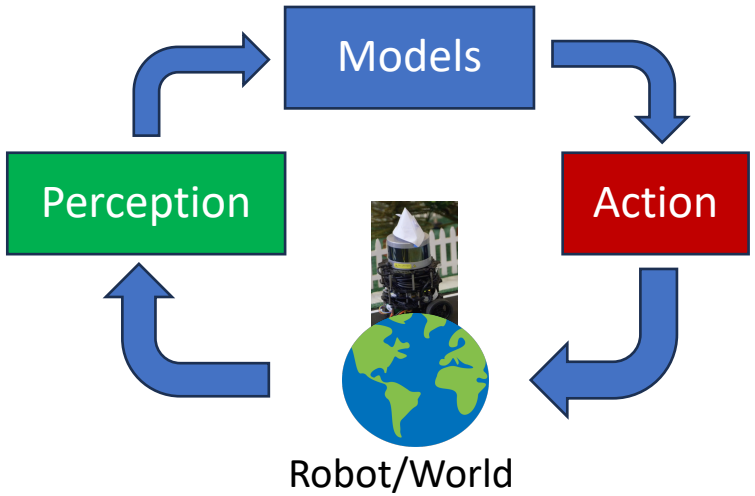

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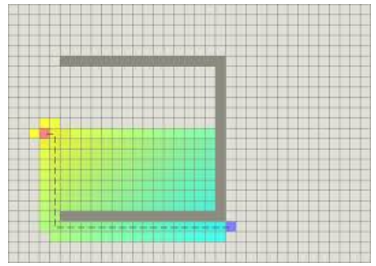

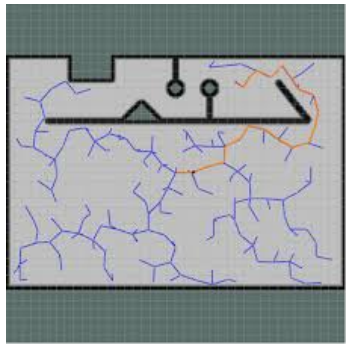

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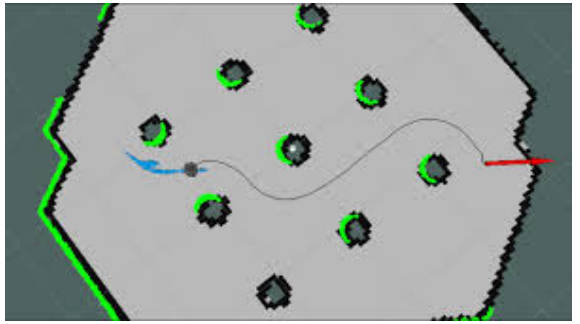

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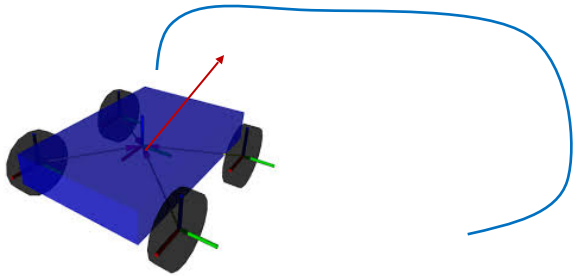

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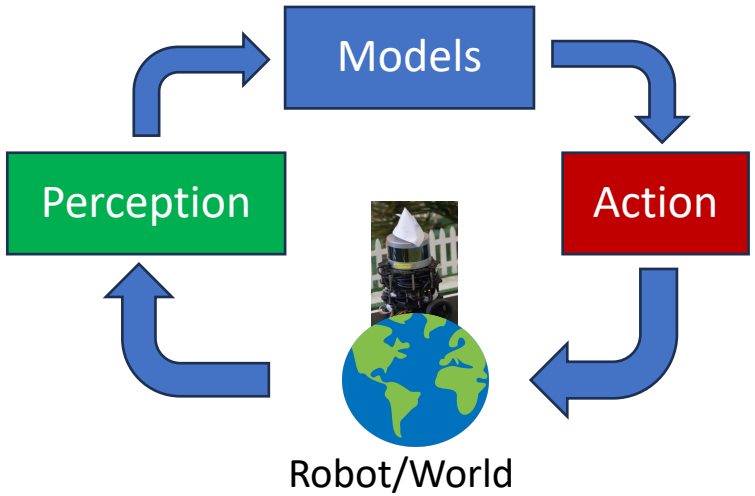

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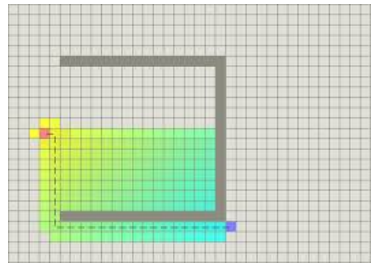

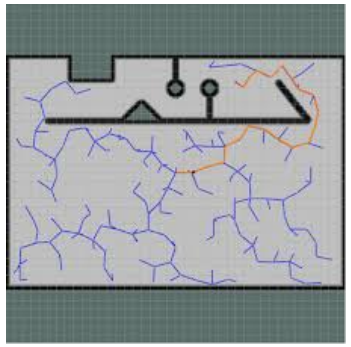

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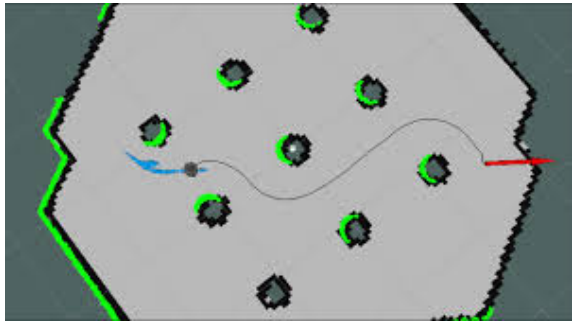

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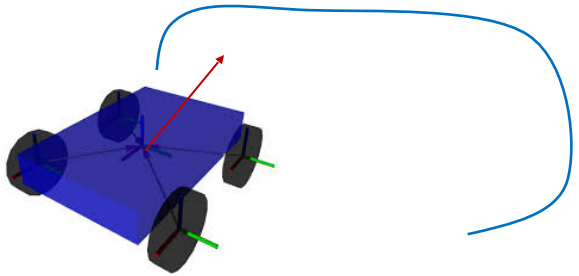

Path Planning

Map representation

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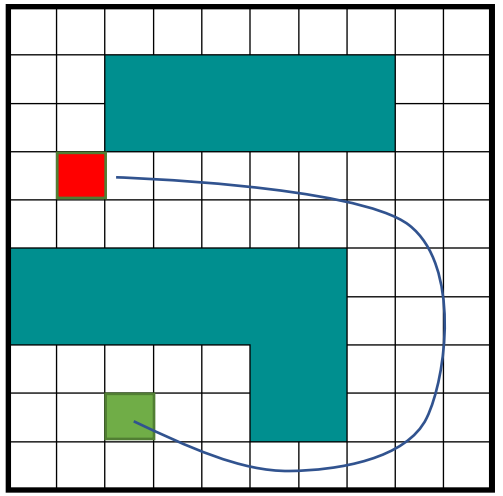

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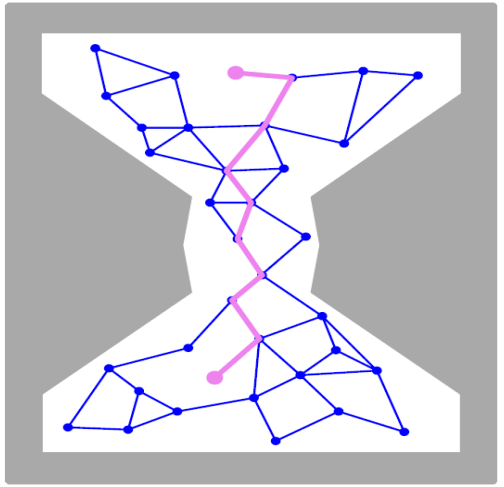

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Graph search planning

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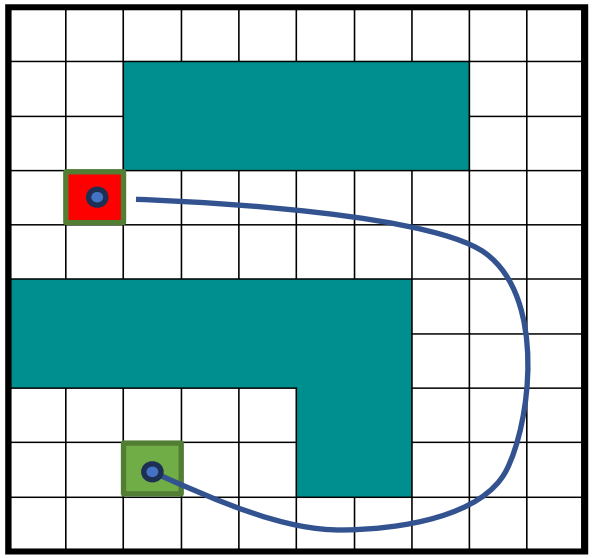
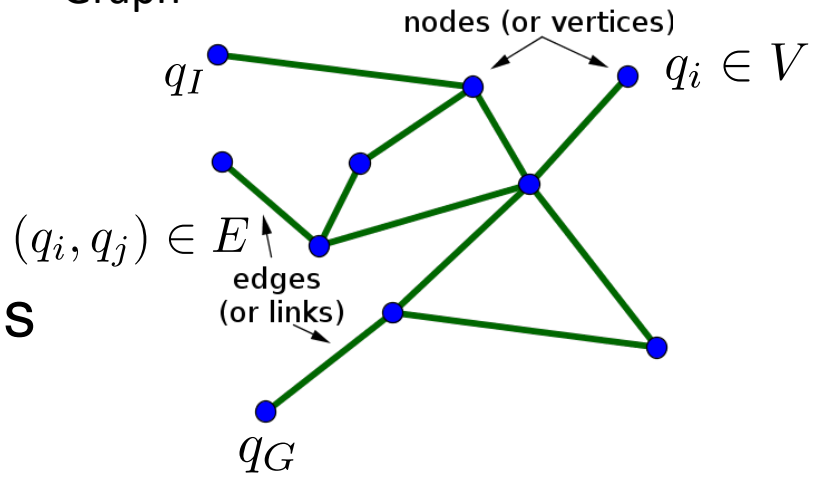


- Discretize robot's **configuration space** into a grid
  - Each grid cell is either free or occupied
  - Robot moves between adjacent free cells
  - **Goal:** find shortest sequence of free cells from start to goal

- Mathematically, this corresponds to pathfinding in a discrete graph $G = (V, E)$
  - Each vertex $q_i \in V$ represents a free cell
  - Edges $(q_i, q_j) \in E$ connect adjacent grid cells

Graph



Graph search algorithms

- Having determined decomposition, how to find “best” path?
- **Label-Correcting Algorithms**: $C(q)$: *cost-to-come* from q_I to q
- Set to ∞ for all nodes, except start node to 0, frontier queue has only start node

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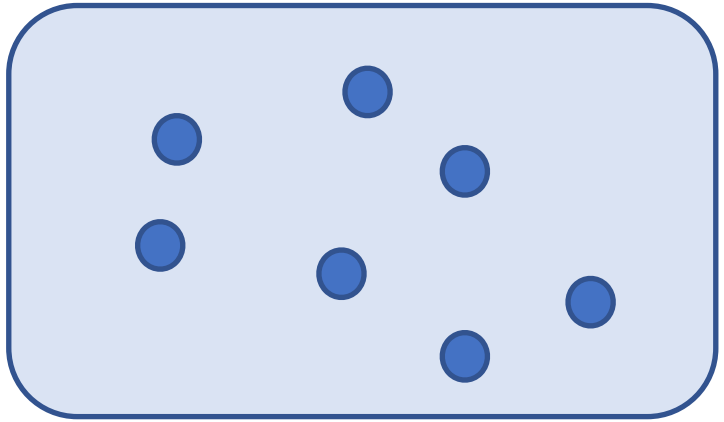

FRONTIER/ALIVE/PRIORITY QUEUE

Graph search algorithms

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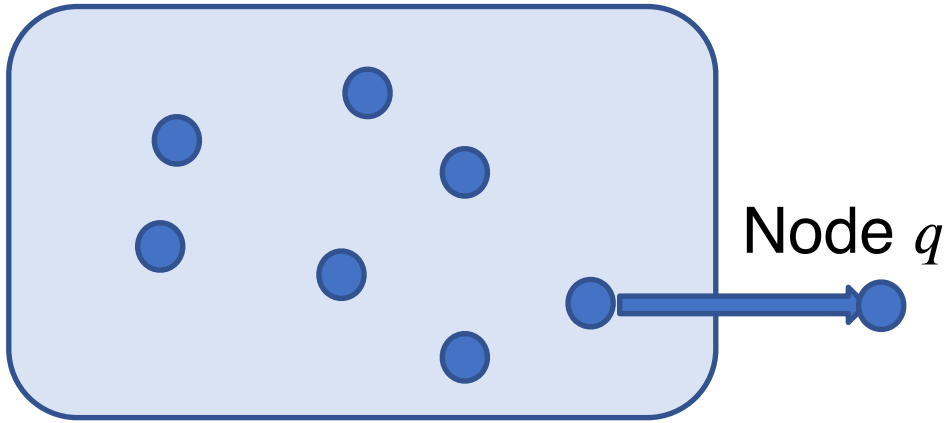

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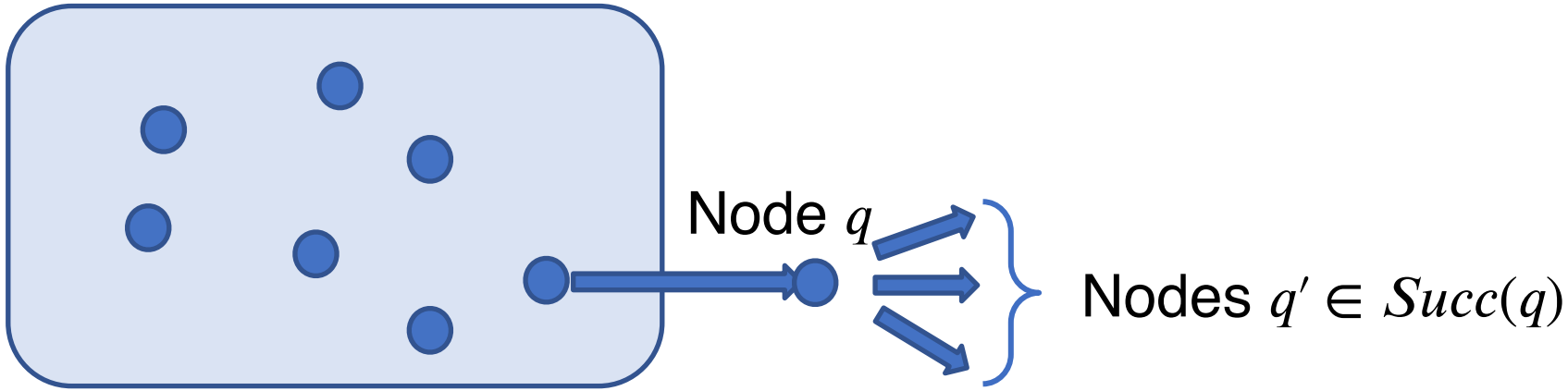

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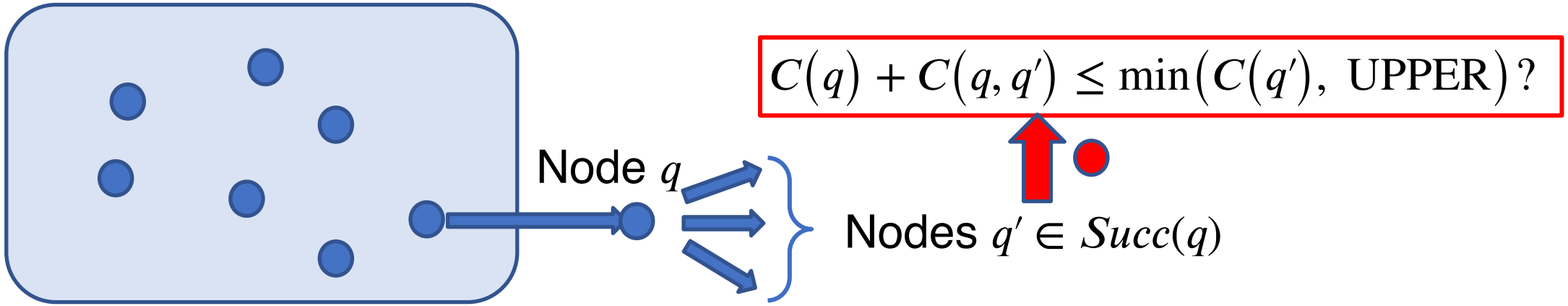

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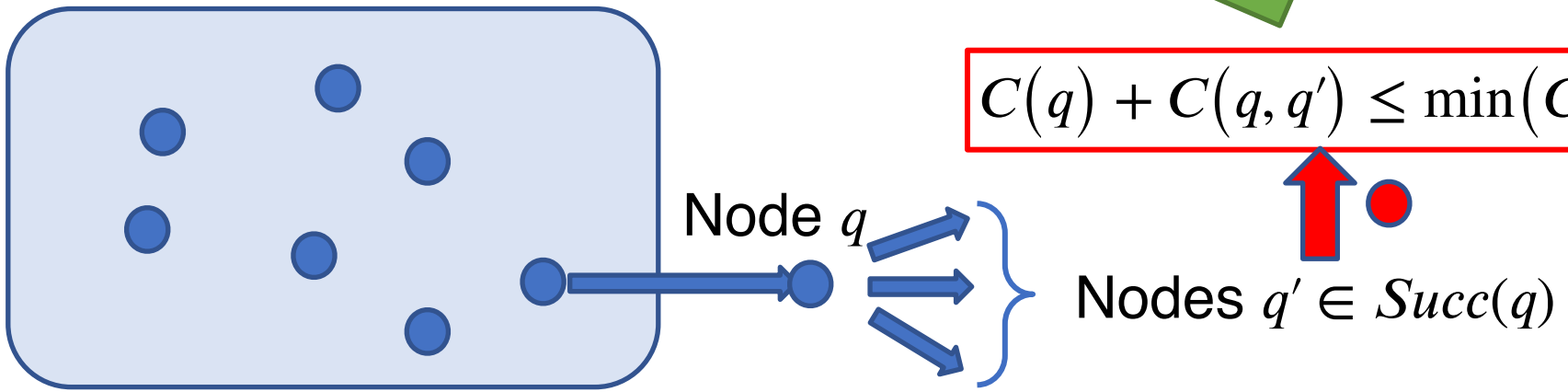
Yes $\Rightarrow C(q') := C(q) + C(q, q')$
Parent $(q') := q$

$C(q) + C(q, q') \leq \min(C(q'),\ \text{UPPER})$?



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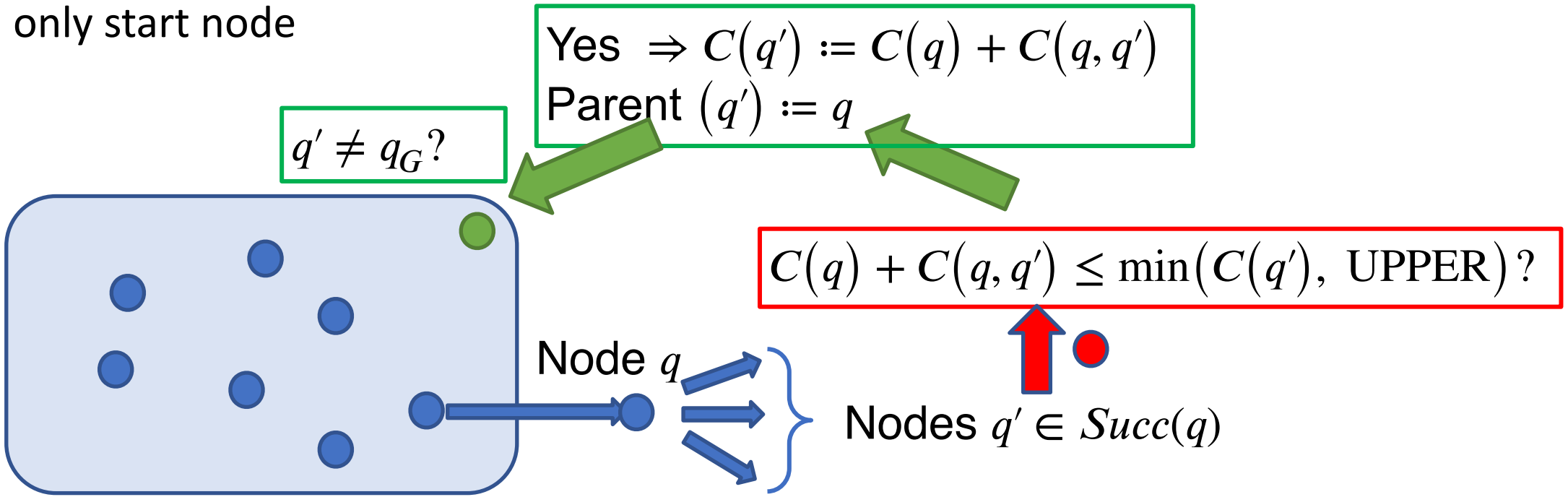

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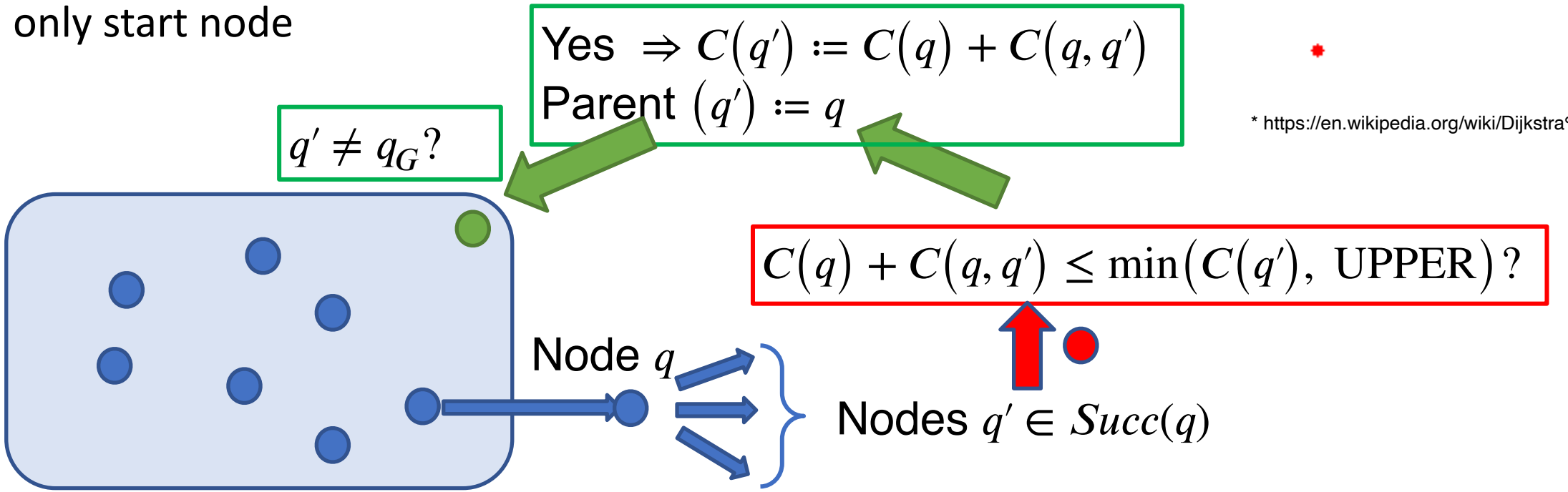


* https://en.wikipedia.org/wiki/Dijkstra%27s_algorithm

FRONTIER/ALIVE/PRIORITY QUEUE

Label correcting algorithm

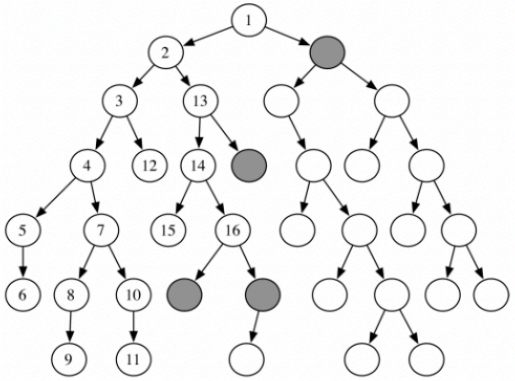
Initialization: set the labels of all nodes to ∞, except for the label of the origin node, which is set to 0, frontier queue has only start node.

Step 1. Remove a node q from frontier queue Q and for each child q' of q, execute step 2

Step 2. If $C(q) + C(q, q') \leq \min(C(q'), \text{ UPPER})$, set $C(q') := C(q) + C(q, q')$ and set q to be the parent of q'. In addition, if $q' \neq q_G$, place q' in the frontier queue if it is not already there, while if $q' = q_G$, set UPPER to the new value $C(q) + C(q, q_G)$

Step 3. If the frontier queue is empty, terminate, else go to step 1

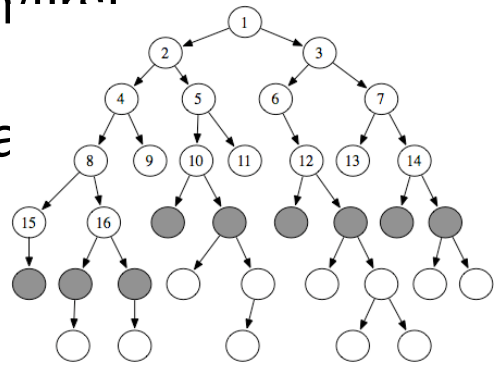
How to prioritize the queue?



Depth-First-Search (DFS): Maintain Q as a **stack** – Last in/first out

- Lower memory requirement (only need to store part of graph)

Breadth-First-Search (BFS, Bellman-Ford): Maintain Q as a **list** – First in/first out



- Update cost for all edges up to current depth before proceeding to greater depth
- Can deal with negative edge (transition) costs

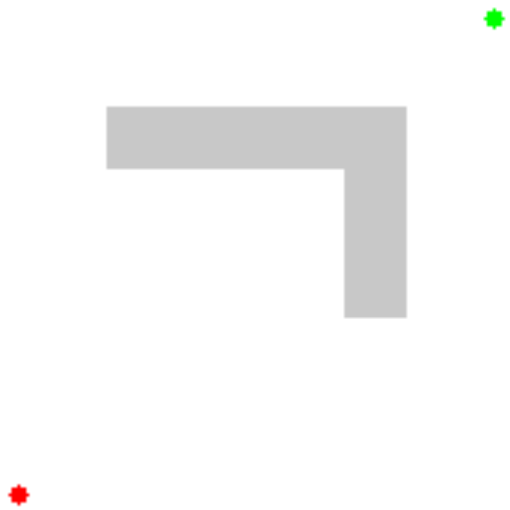
Best-First (BF, Dijkstra): Greedily select next q: $q = \operatorname{argmin}_{q \in Q} C(q)$

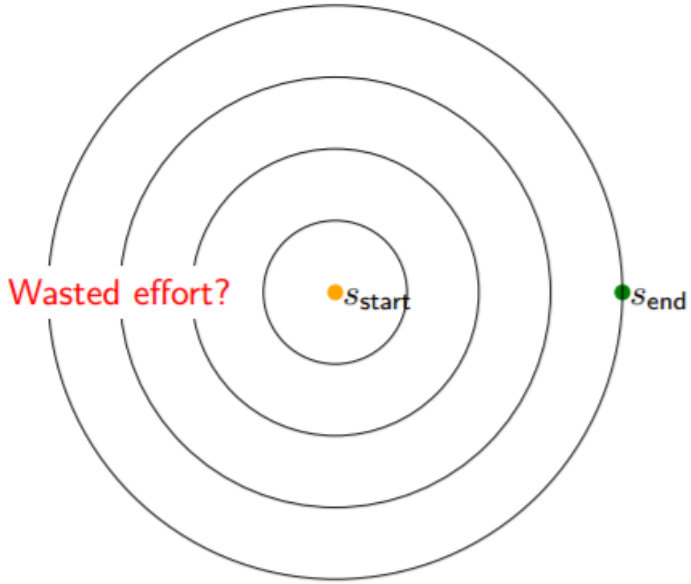
- Node will enter the frontier queue at most *once*
- Requires costs to be non-negative

Correctness and improvements

Theorem

If a feasible path exists from q_I to q_G, then algorithm terminates in finite time with $C(q_G)$ equal to the optimal cost of traversal, $C^*(q_G)$.





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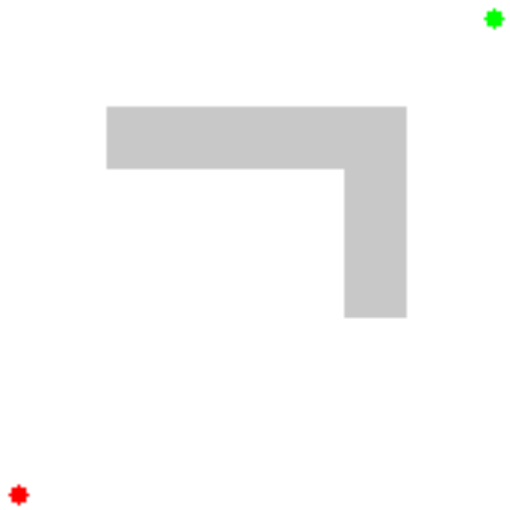

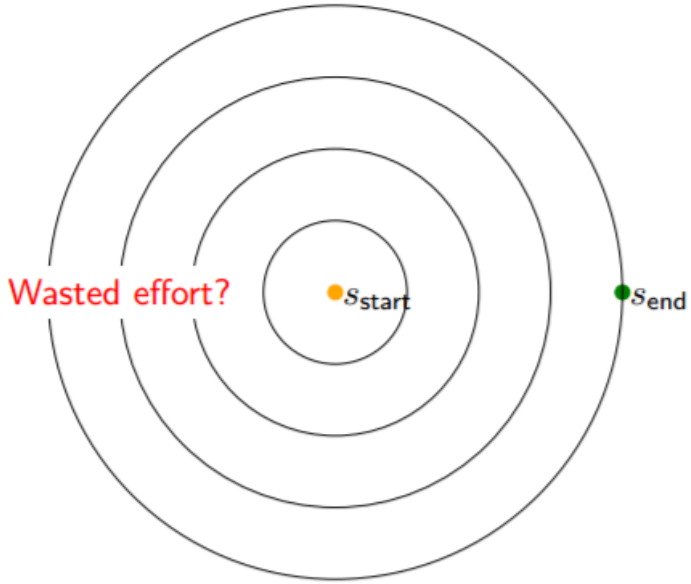

A*: Improving Dijkstra (1968)

- Dijkstra orders by optimal "*cost-to-come*"
- Faster results if order by "*cost-to-come*"*+ (approximate) "cost-to-go"*
- Take the node with lowest C(q) + h(q) from frontier set, where $h(q)$ is *a* heuristic for optimal cost-to-go (specifically, a positive *underestimate*)
- When testing to add neighbor into frontier set, use stronger

$$C(q) + C(q, q') + h(q') \leq \text{UPPER}$$

Instead of $C(q) + C(q, q') \leq \text{UPPER}$

- In this way, fewer nodes will be placed in the frontier queue
- This modification still guarantees that the algorithm will terminate with a shortest path

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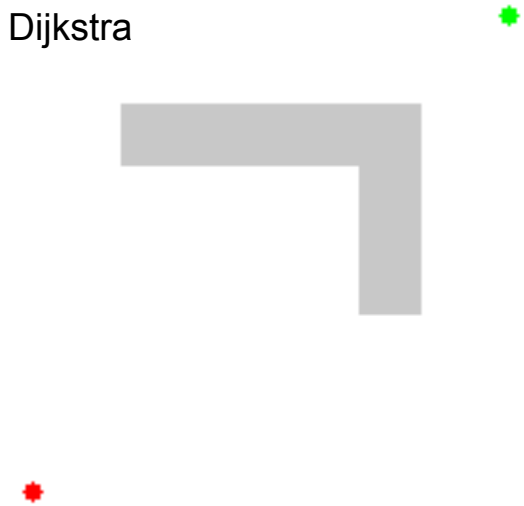

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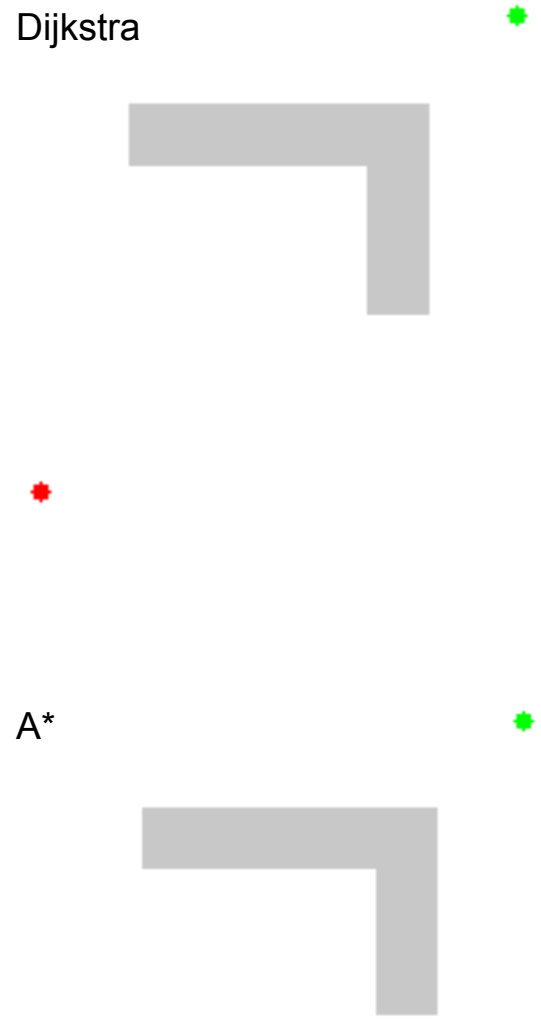

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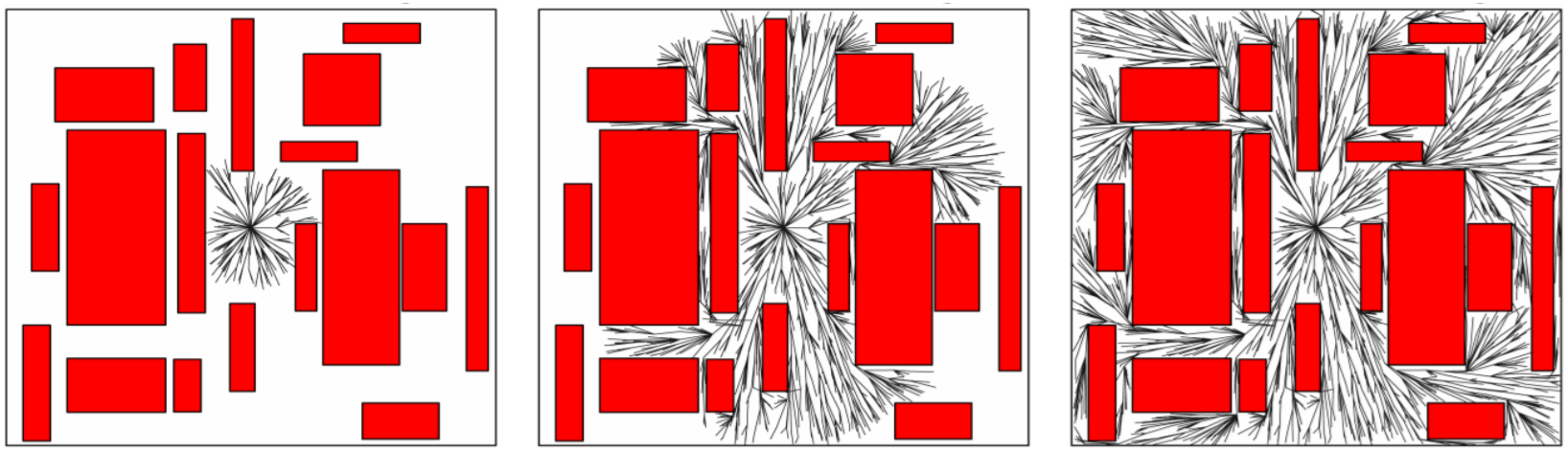
Graph search approaches: summary

- Pros:
  - Simple and easy to use
  - Fast (for some problems)
- Cons:
  - Resolution dependent
    - Not guaranteed to find solution if grid resolution is not small enough
  - Limited to simple robots
    - C-space grid size is exponential in the number of DOFs

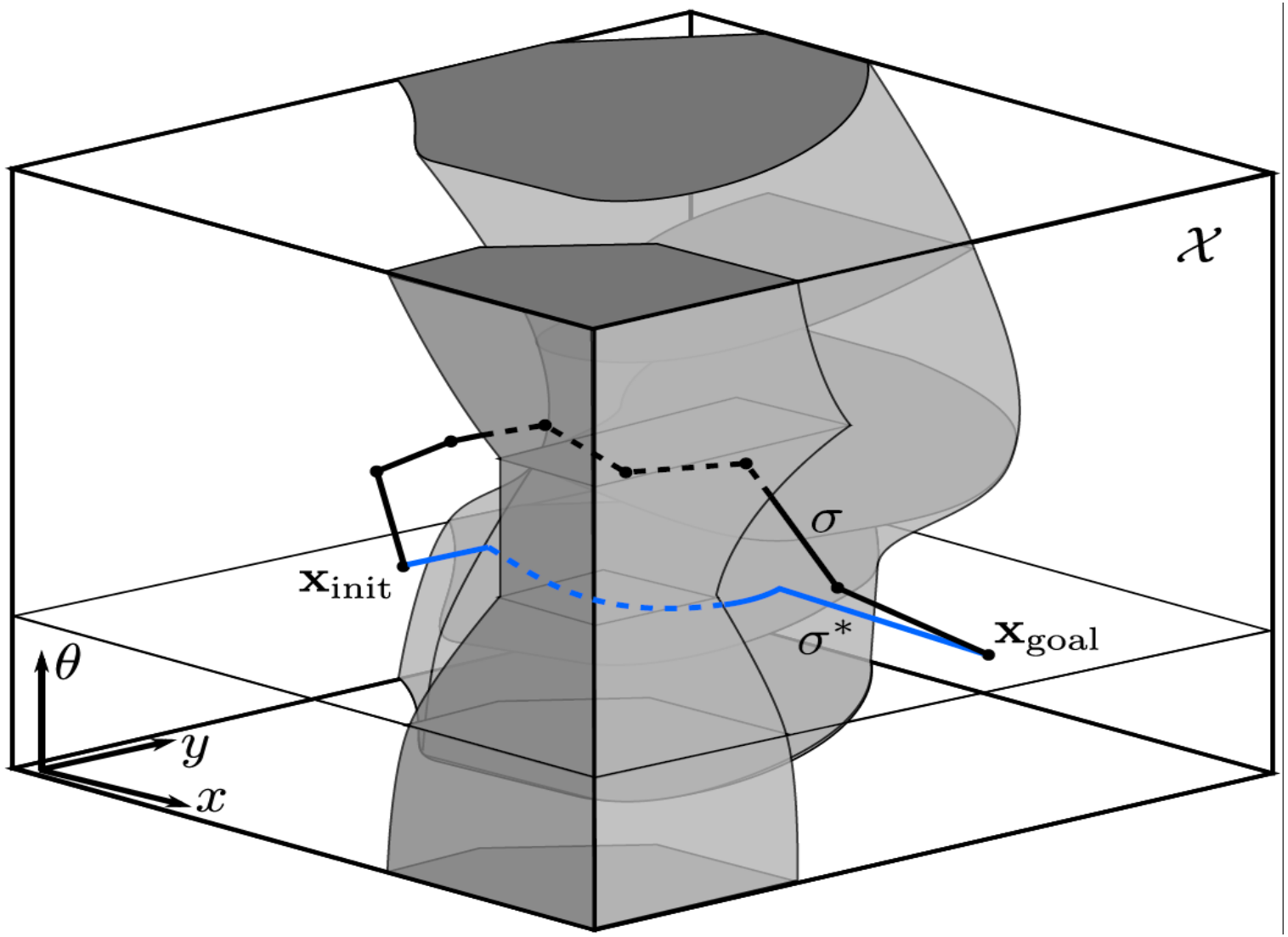
Sampling based planning

Build a **random graph** of pose waypoints and edges between them. Find shortest path in this graph either: after construction (PRM), or during construction (RRT, RRT*).



Karaman, Frizzoli, IJRR 2011, RRT*.

Recall, C-Space



Sampling-based planning

Challenges of explicit C-space computation motivated the development of sampling-based approaches

- Abandon the idea of explicitly characterizing C_{free} and C_{obs}
- Instead, capture the structure of C by random sampling
- Use a black-box component (collision checker) to determine which random configurations lie in C_{free}
- Use such a probing scheme to build a roadmap (a graph of possible paths) and then plan a path

History of sampling-based planning

Traditionally, two major approaches:

- Probabilistic Roadmap (PRM): graph-based
  - Multi-query planner, i.e., designed to solve multiple path queries on the same scenario
  - Original version: [Kavraki et al., ‘96]
  - “Lazy” version: [Bohlin & Kavraki, ‘00]
  - Dynamic version: [Jaillet & T. Simeon, ‘04]
  - Asymptotically optimal version: [Karaman & Frazzoli, '11]
- Rapidly-exploring Random Trees (RRT): tree-based
  - Single-query planner
  - Original version: [LaValle & Kuffner, ‘01]
  - RDT: [LaValle, ‘06]
  - SRT: [Plaku et al., ‘05]
  - Asymptotically optimal version [Karaman & Frazzoli, '11]

Probabilistic roadmaps (PRM)

A multi-query planner, which generates a roadmap (**graph**) G, embedded in the free space

Preprocessing step:

1. Sample a collection of n configurations X_n; discard configurations leading to collisions
2. Draw an edge between each pair of samples $x, x' \in X_n$ such that $\|x - x'\| \leq r$
3. And straight-line path between x and x' is collision free

Given a query $s, t \in C_{free}$, connect them to G and find a path on the

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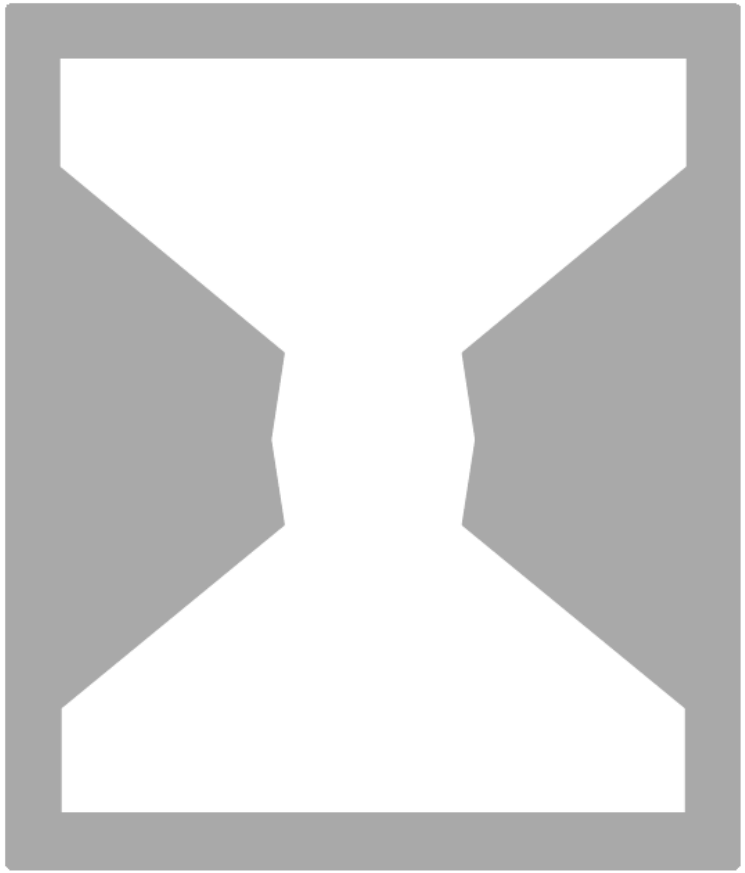

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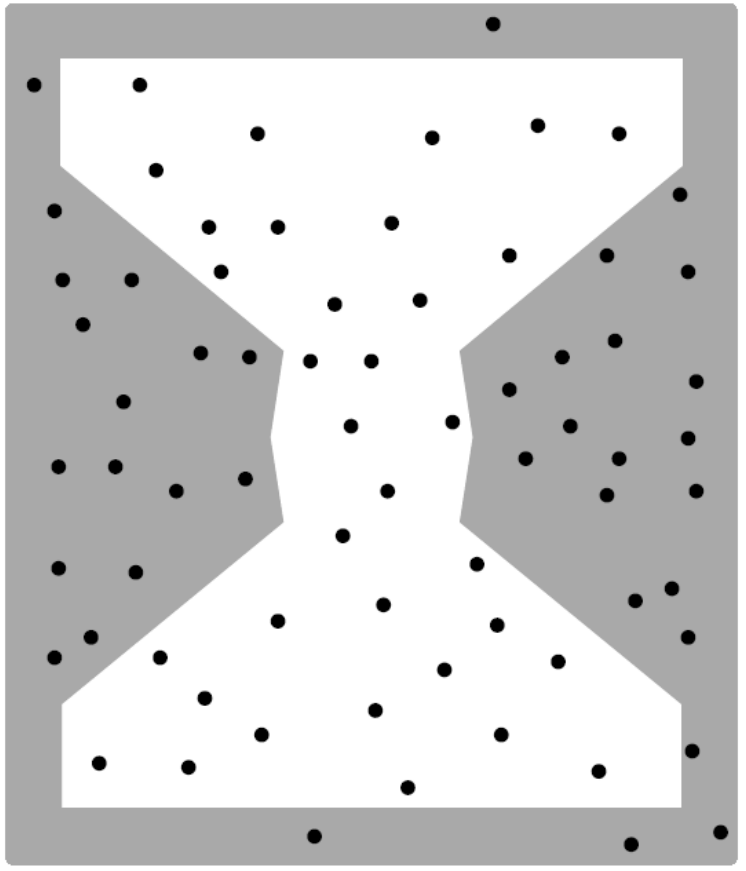

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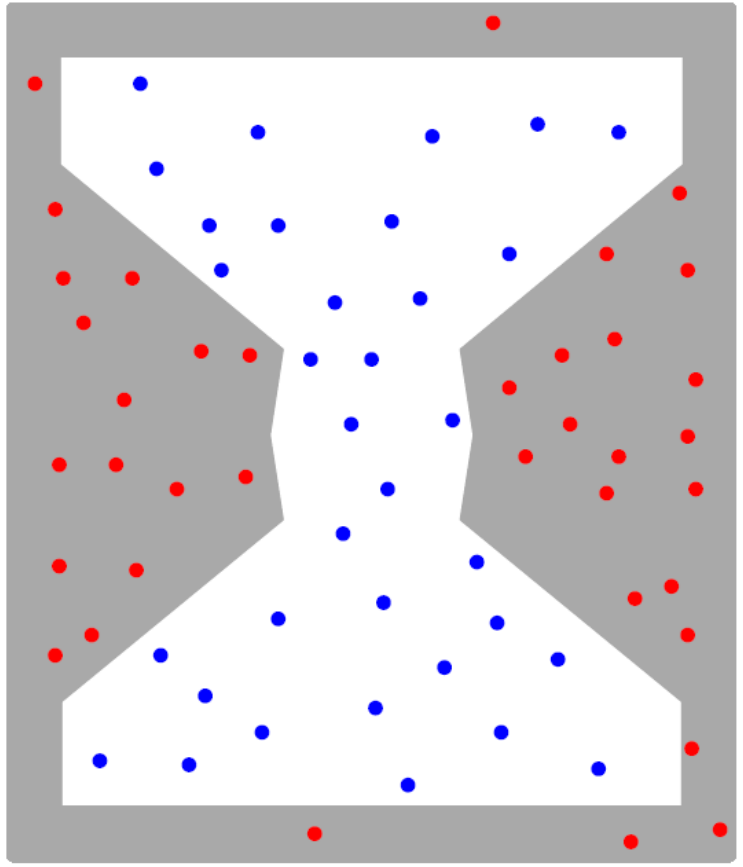

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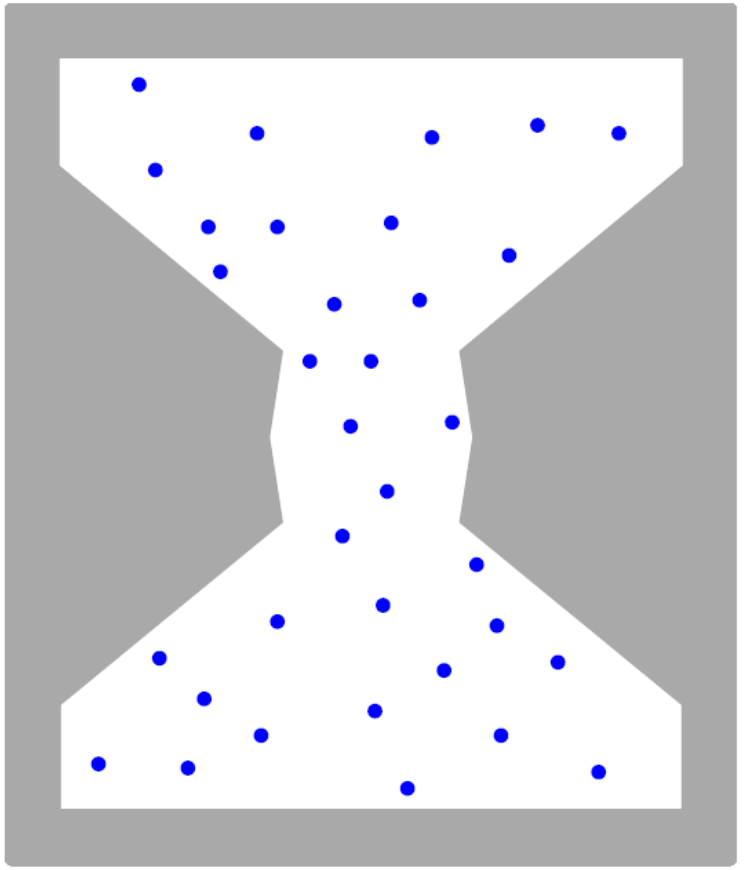

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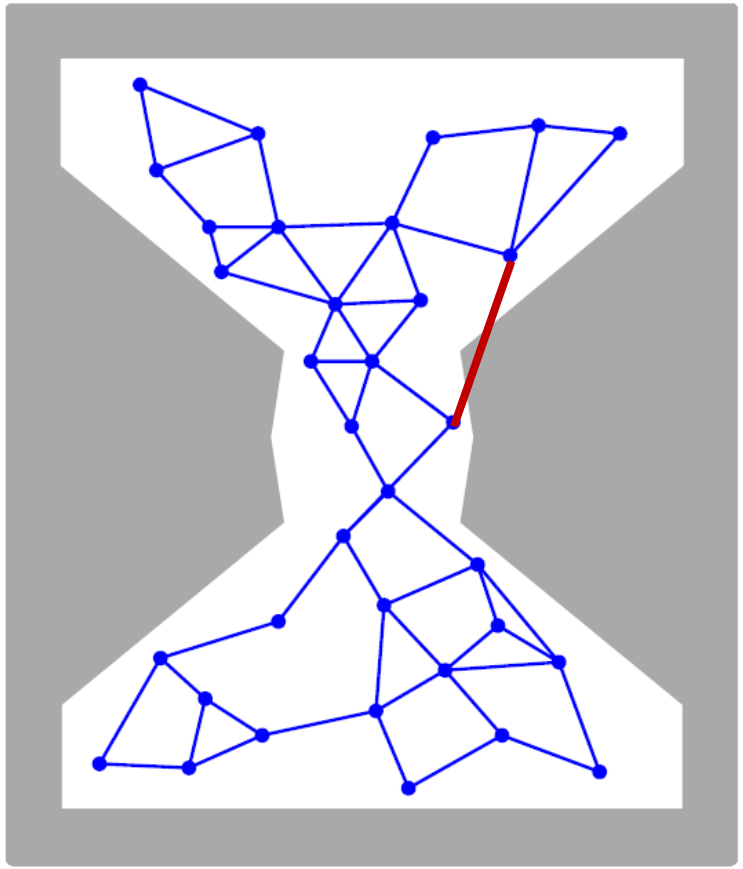

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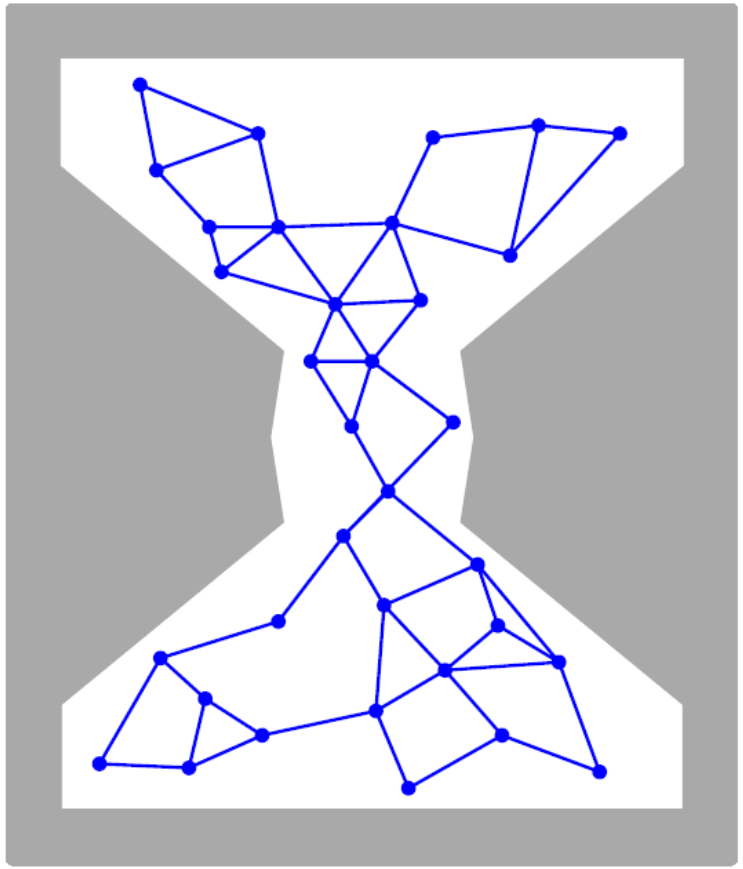

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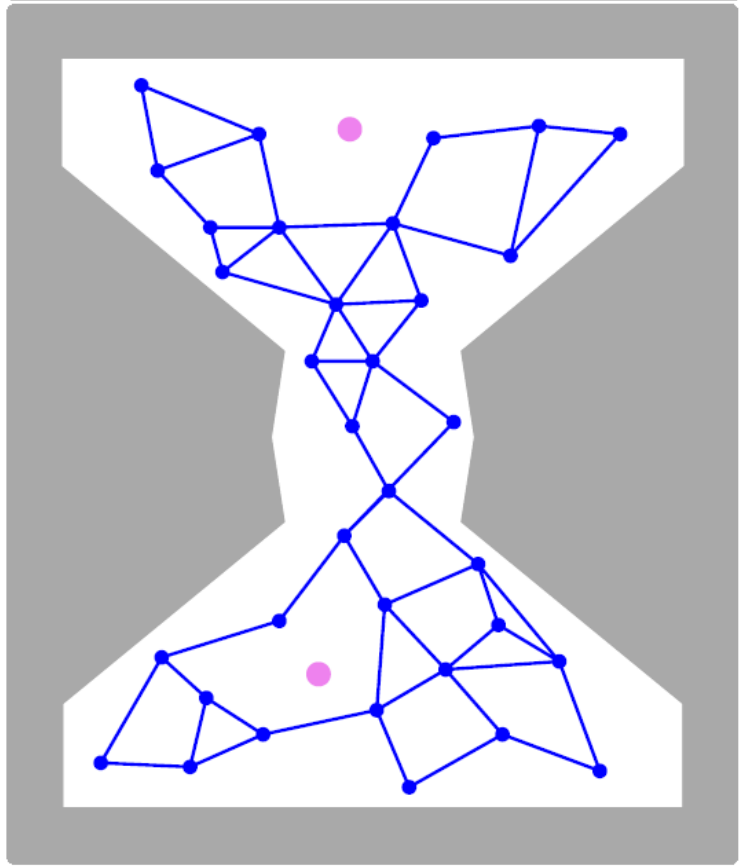

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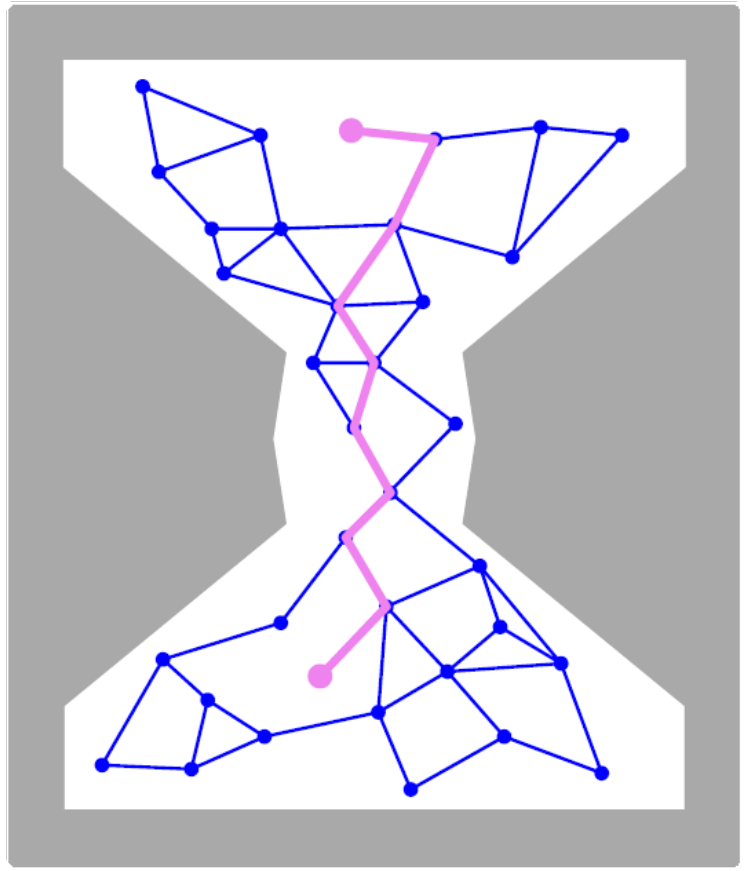

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Rapidly-exploring random trees (RRT)

A single-query planner, which grows a **tree** $T = (V, E)$, rooted at the start configuration s, embedded in C_{free}

Algorithm works in n iterations:

1. Sample configuration x_{rand}
2. Find nearest vertex x_{near} in T to x_{rand}
3. Generate configuration x_{new} in direction of x_{rand} from x_{near}, such that $\|x_{near} - x_{new}\| \leq \eta$ and $\{x \mid \alpha x_{near} + (1-\alpha)x_{new}, \alpha \in [0,1]\} \in C_{free}$
4. Update tree: $T = (V \cup x_{new}, E \cup (x_{near}, x_{new}))$

Occasionally, set x_{rand} to be the target vertex t; terminate when $x_{new} = t$

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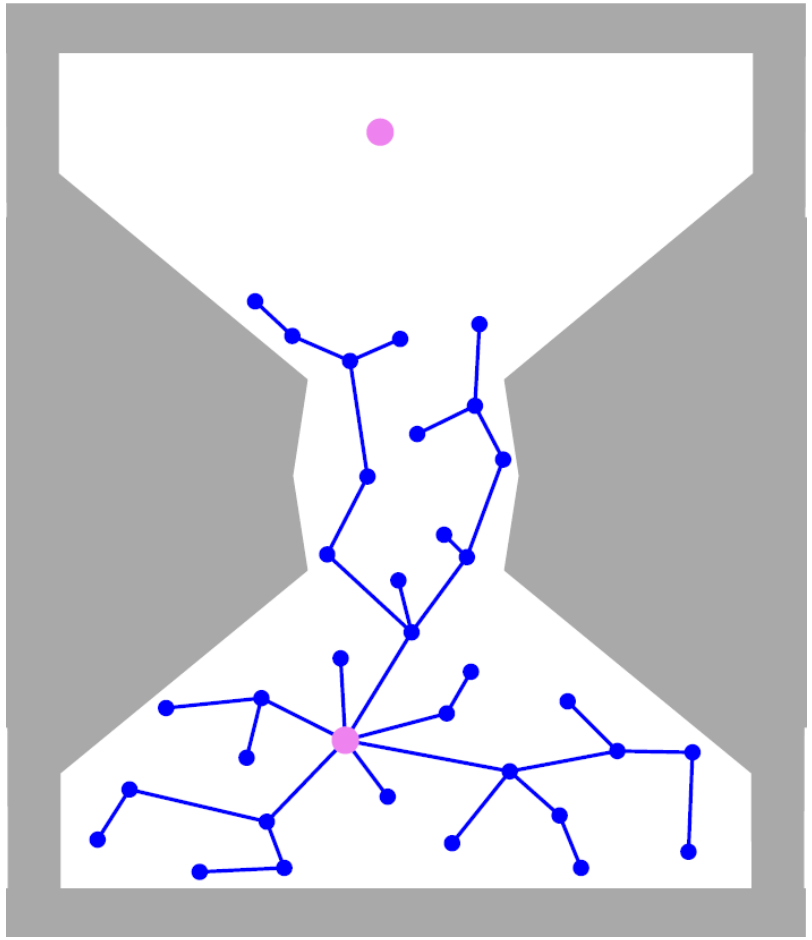

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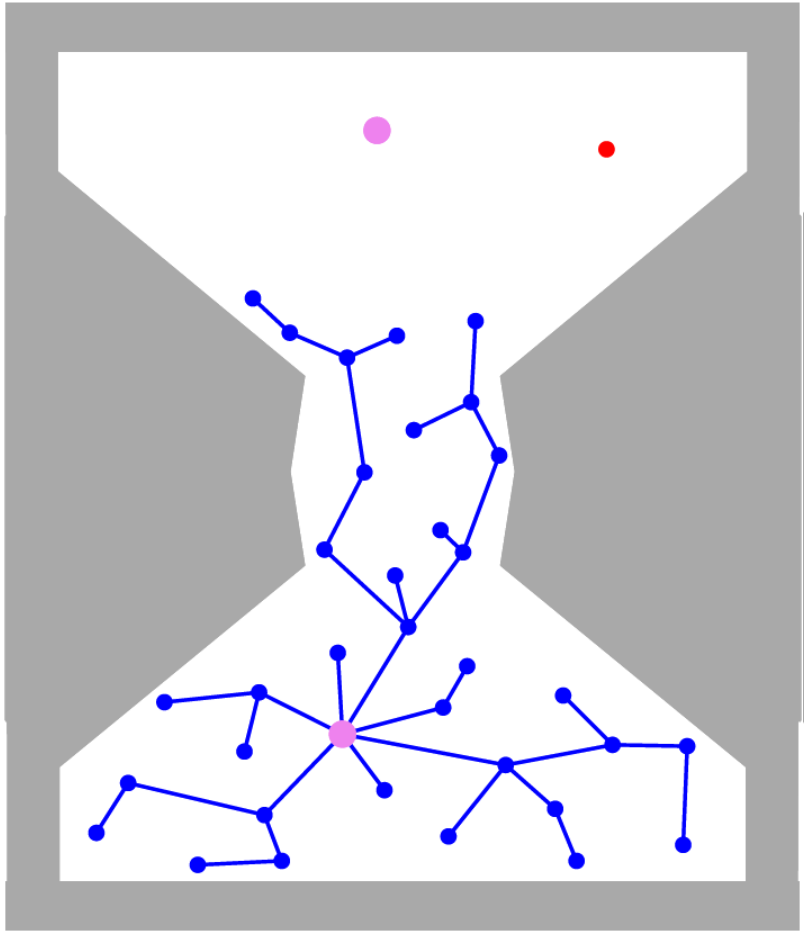

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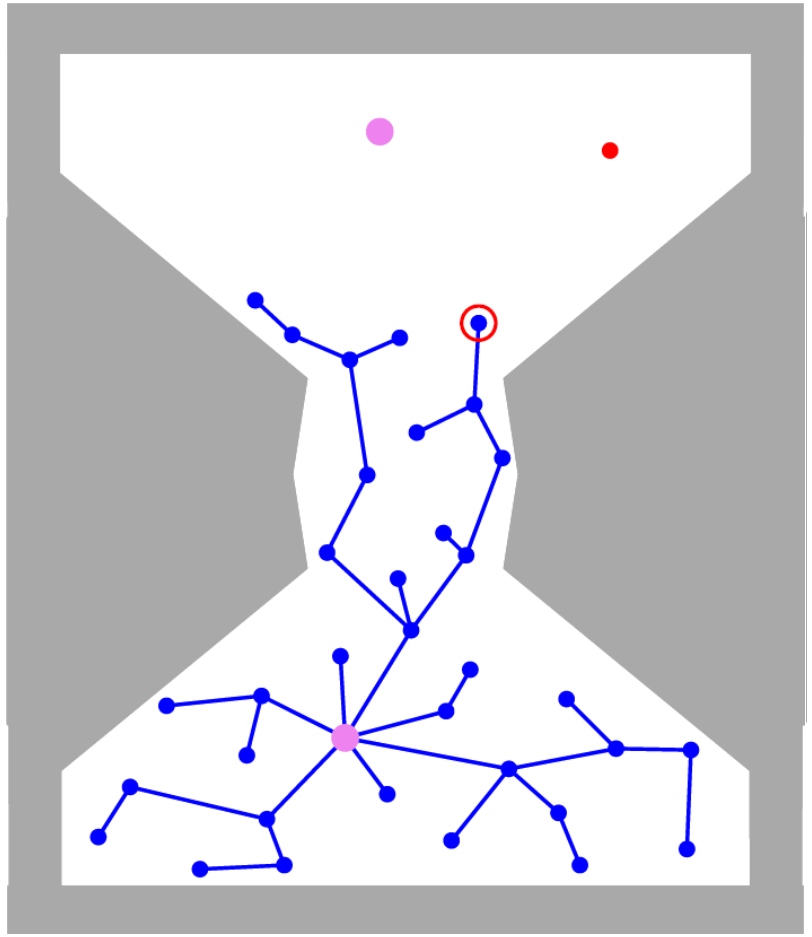

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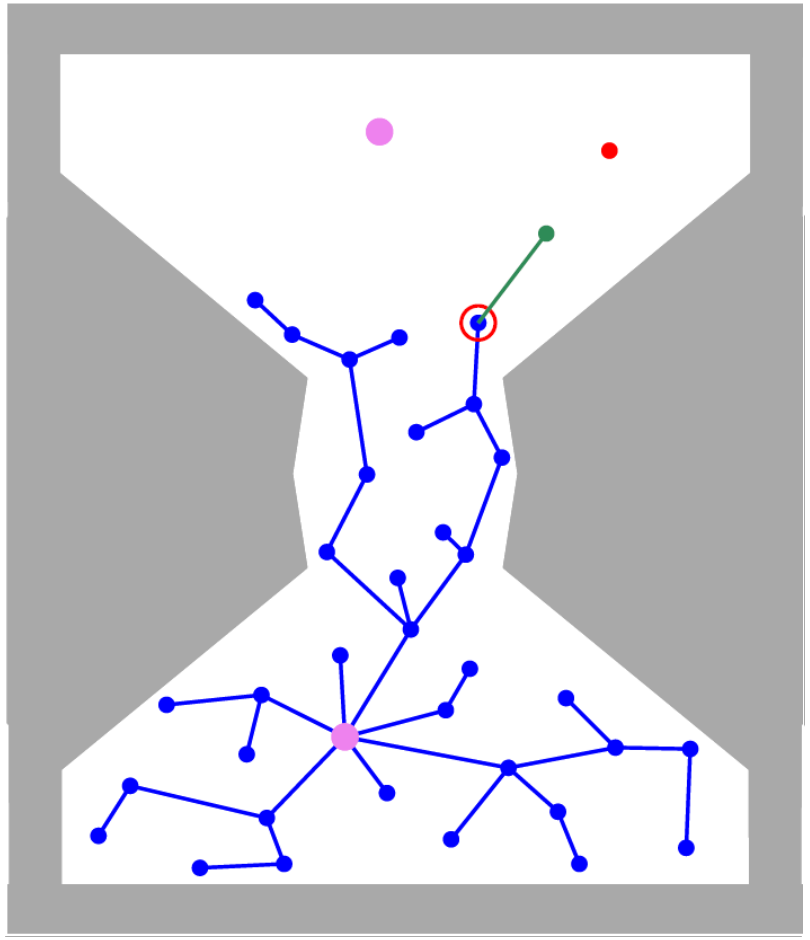

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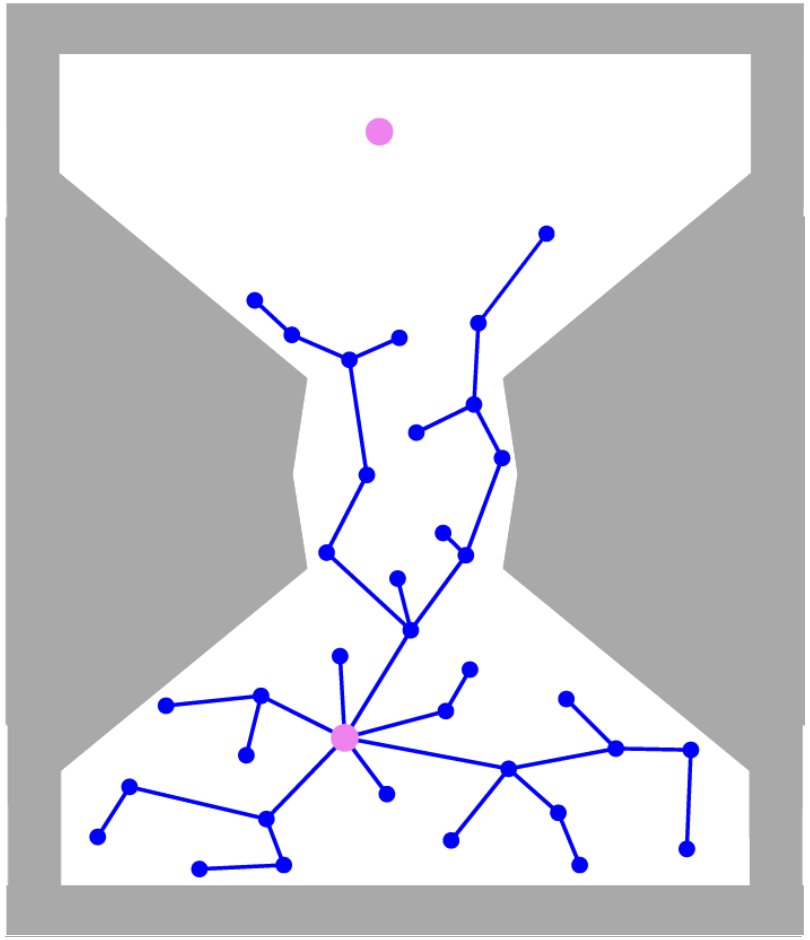

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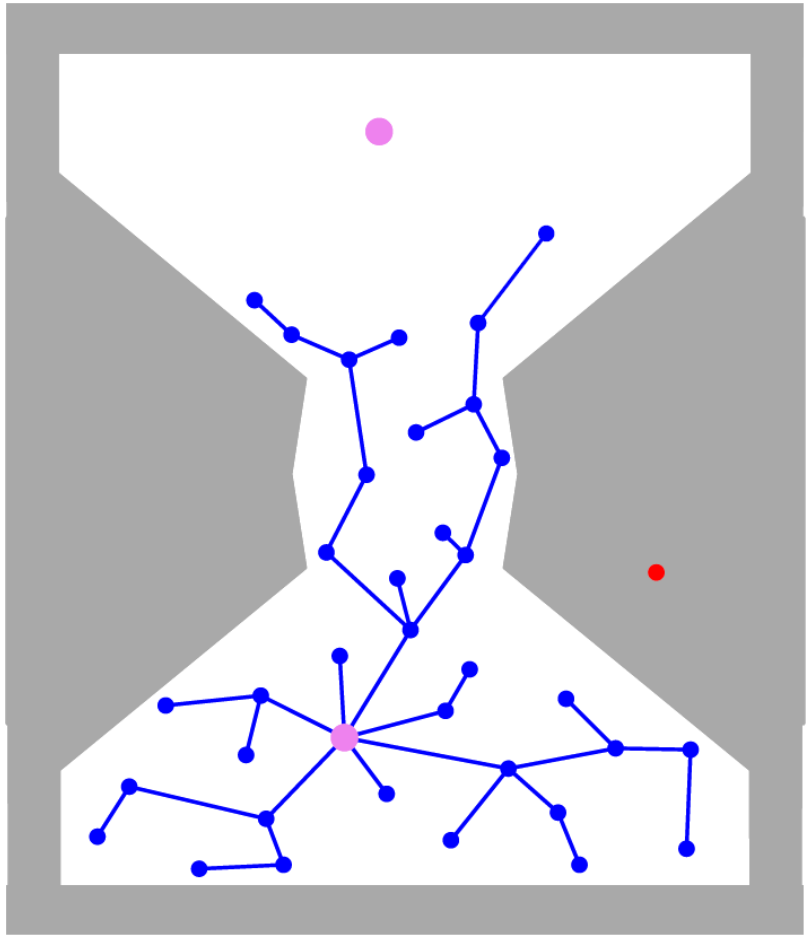

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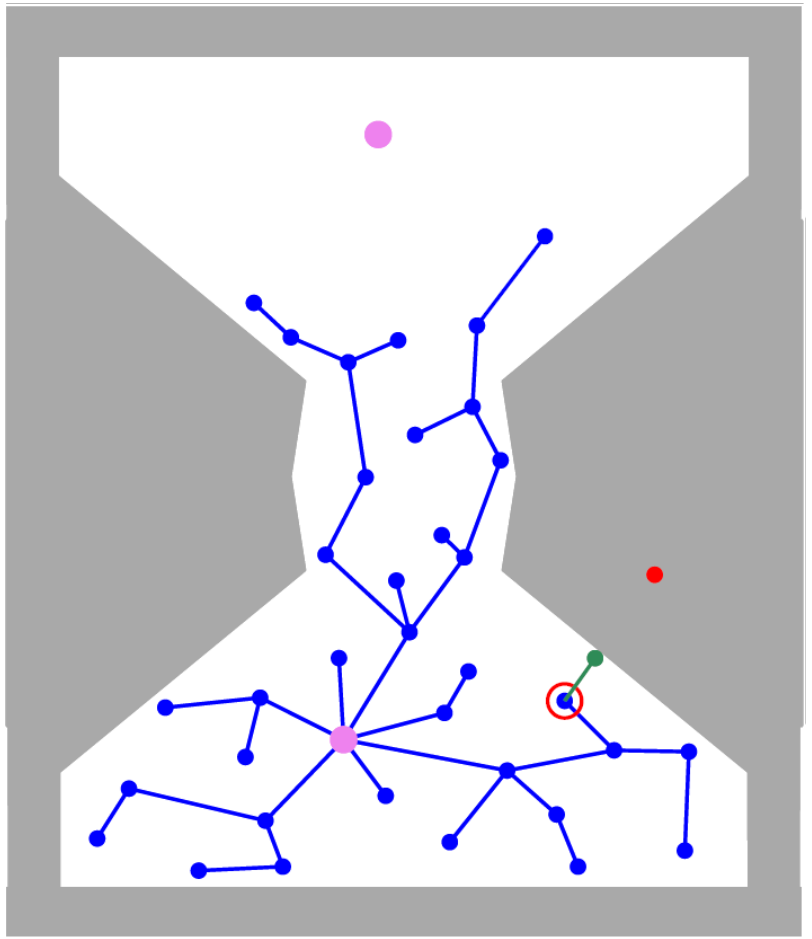

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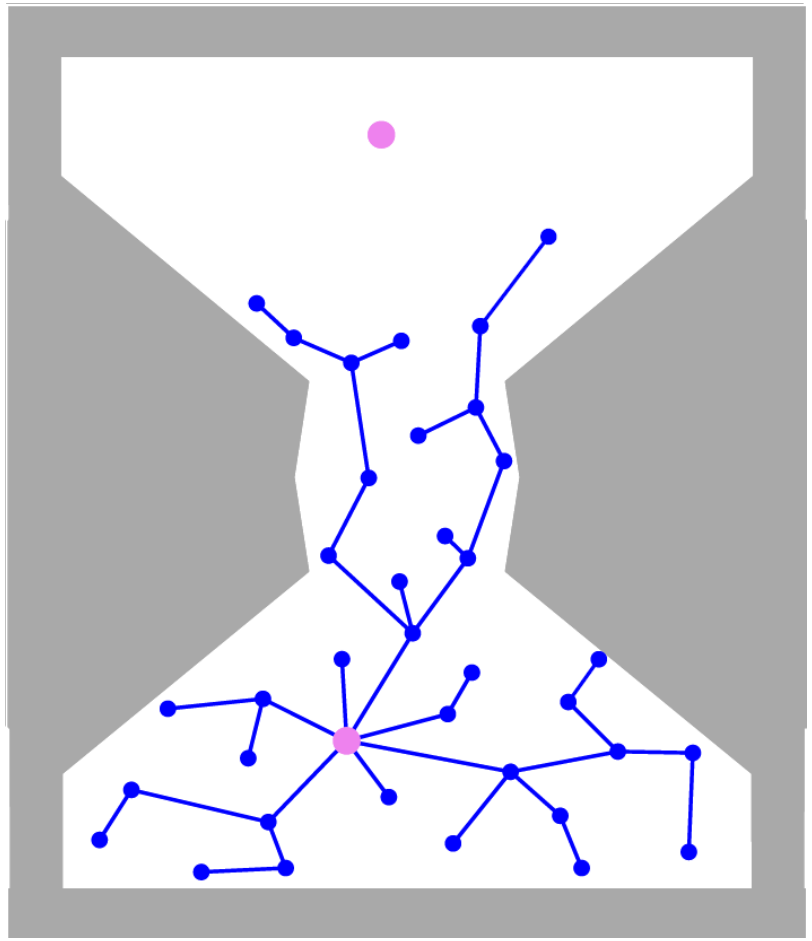

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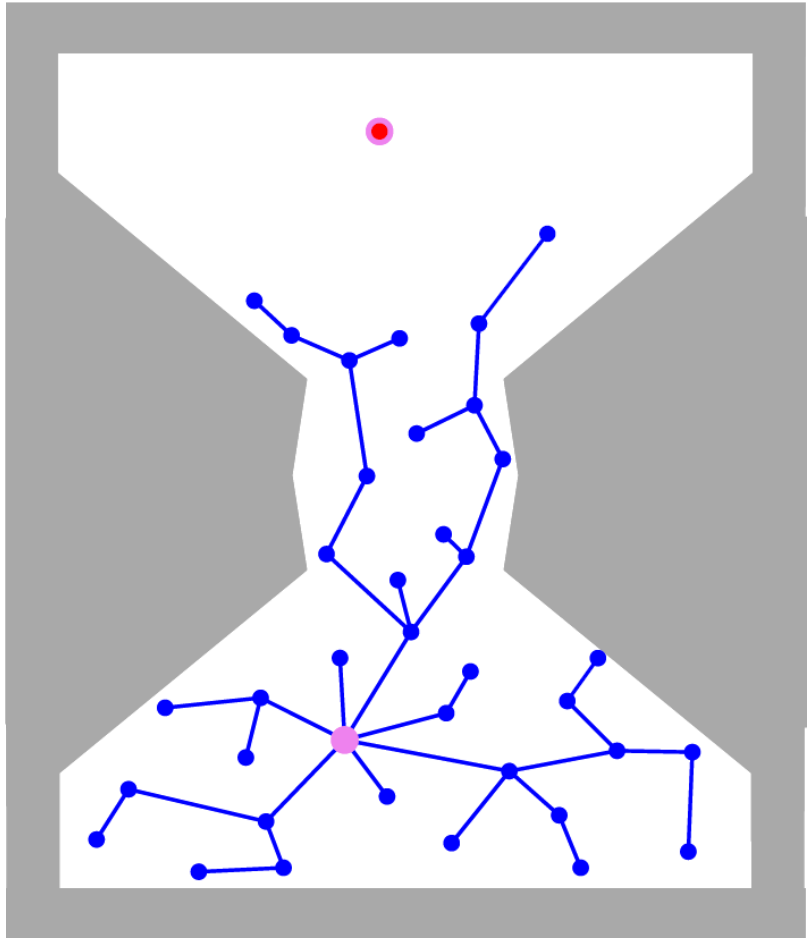

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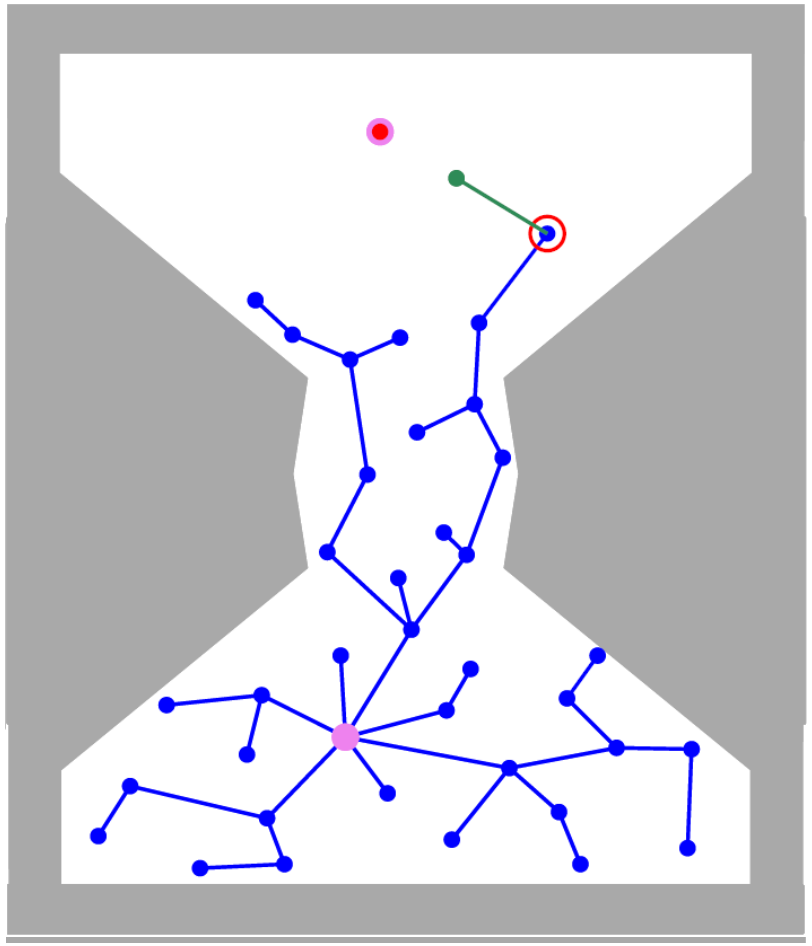

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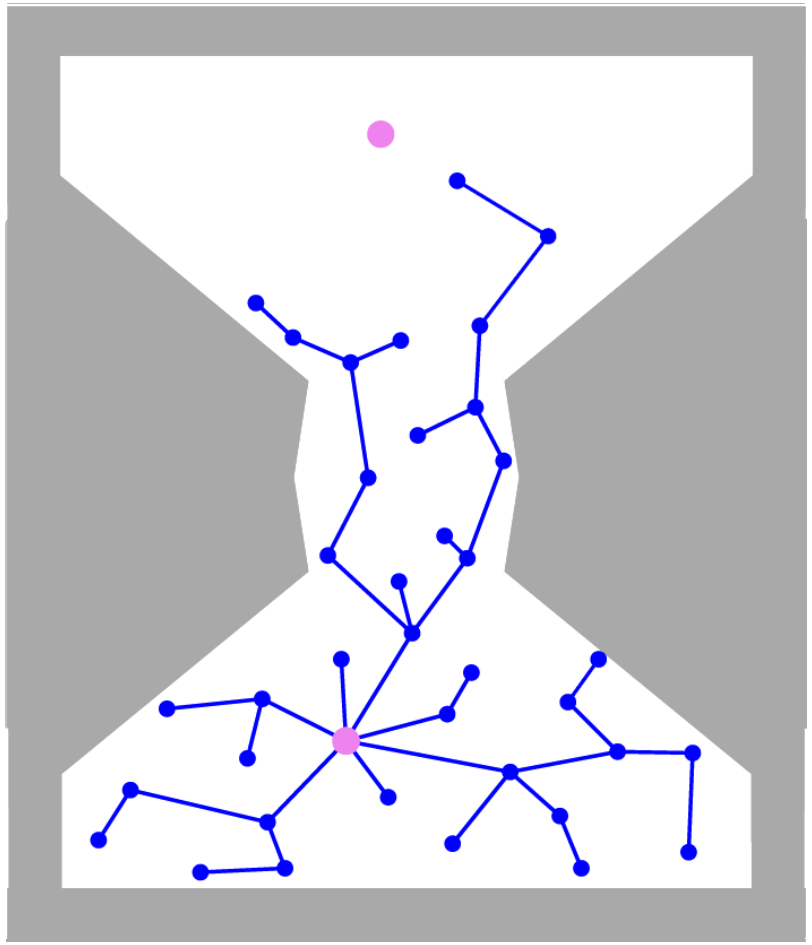
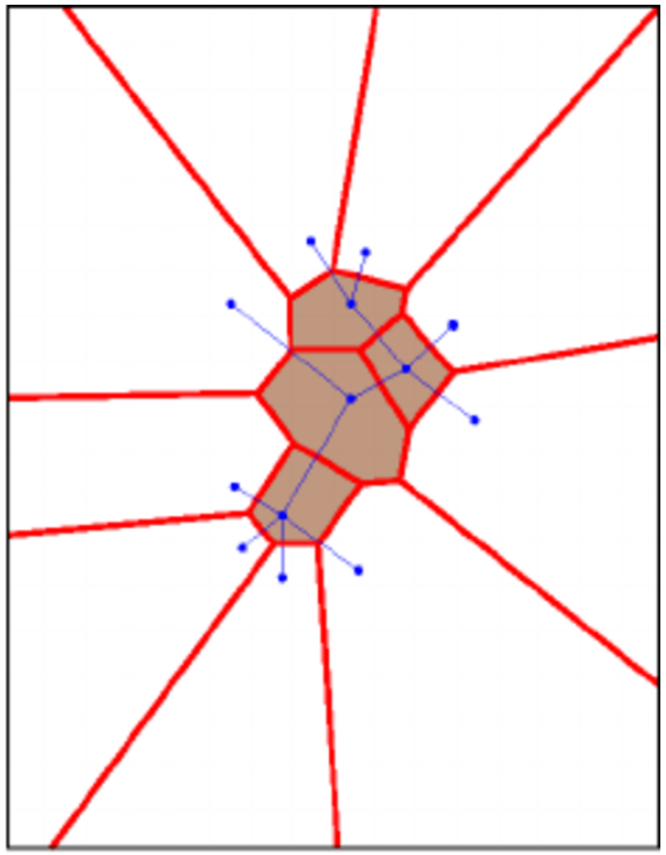


Rapidly-exploring random trees (RRT)

- RRT is known to work quite well in practice
- Its performance can be attributed to its Voronoi bias:
  - Consider a Voronoi diagram with respect to the vertices of the tree
  - For each vertex, its Voronoi cell consists of all points that are closer to that vertex than to any other
  - Vertices on the frontier of the tree have larger Voronoi cells – hence sampling in those regions is more likely



Theoretical guarantees: probabilistic completeness

Question: how large should the number of samples n be? We can say something about the asymptotic behavior:

Kavraki et al. 96: For PRM, with $r = \text{const}$, the probability of finding a path if one exists approaches 1 as $n \to \infty$.

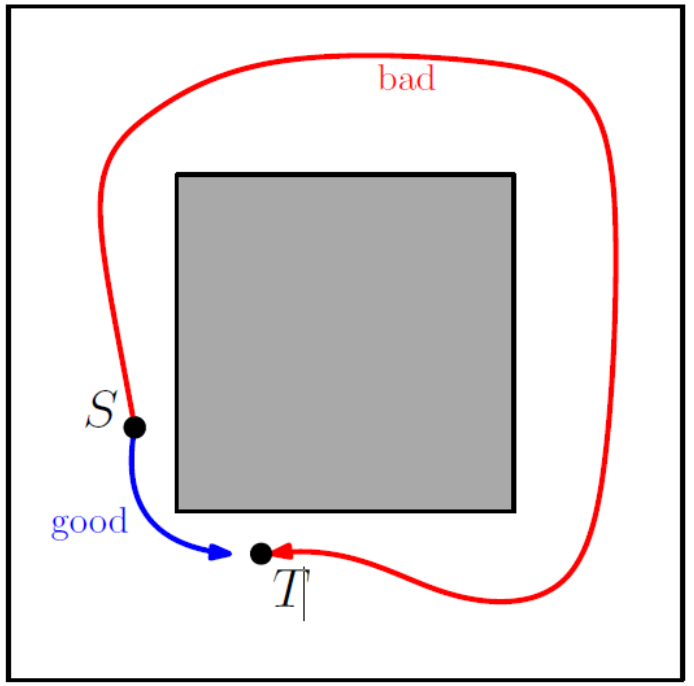
LaValle, 98; Kleinbort et al., 18: For RRT, the probability of finding a path if one exists approaches 1 as $n \to \infty$

* hypercube $[0,1]^d$, with $2 \leq d < \infty$

Theoretical guarantees: quality

Question: what can be said about the quality of the returned solution for PRM and RRT, in terms of length, energy, etc.?

- Nechushtan et al. (2011) and Karaman and Frazzoli (2011) proved that RRT can produce arbitrarily-bad paths with non-negligible probability: for example, RRT might prefer to take the long (red) way.
- This is because it does not **revise** connections with more efficient choices that might be found later (recall Dijkstra's, A*).

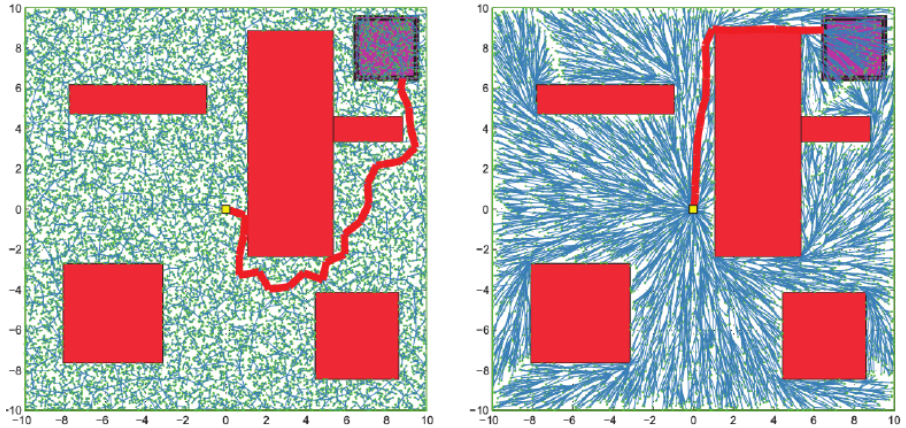


Theoretical guarantees: quality

Karaman and Frazzoli in 2011 provided the first rigorous study of optimality in sampling-based planners:

Theorem: The cost of the solution returned by PRM converges, as $n \to \infty$, to the optimum, when $r_n = \gamma\left(\frac{\log n}{n}\right)^{\frac{1}{d}}$, where γ only depends on d

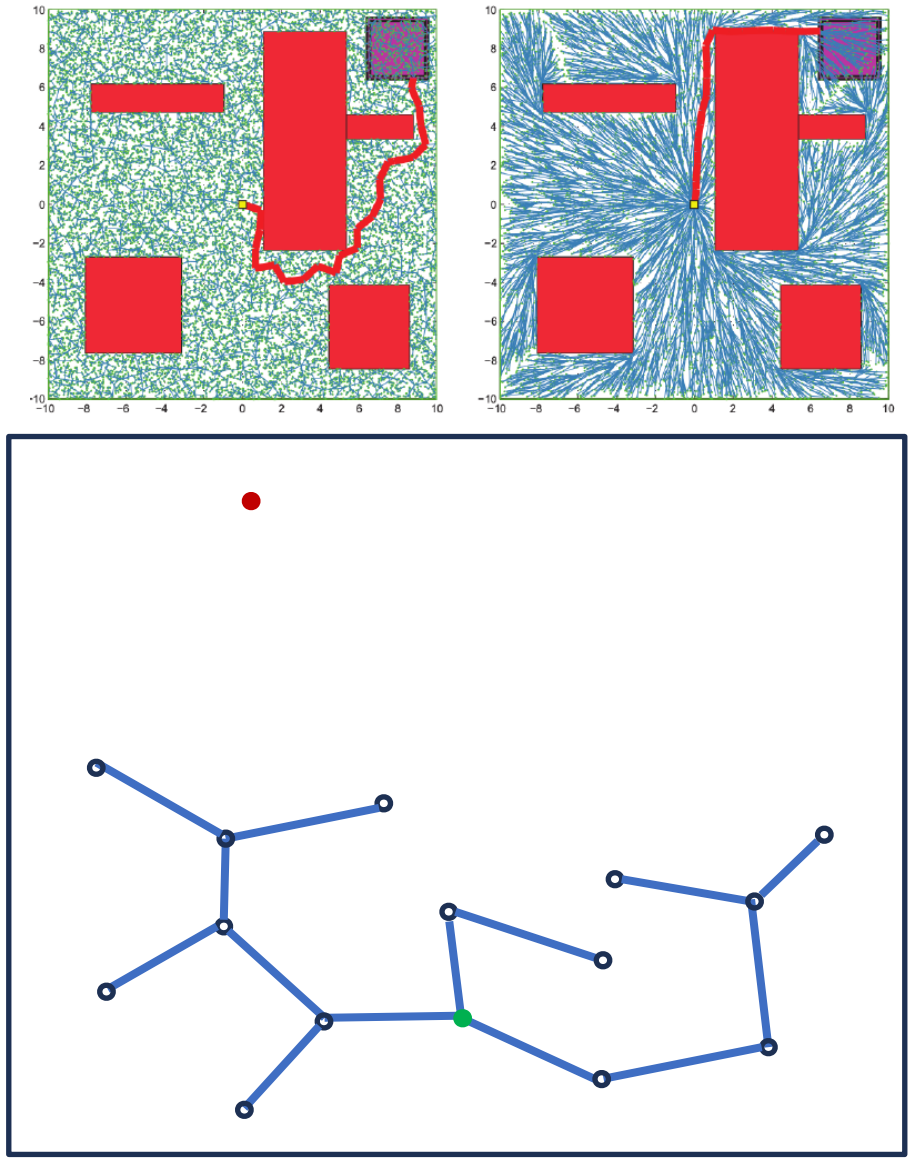
- KF11 also introduced an asymptotically optimal variant of RRT called RRT* (right)
- Introduced **re-wiring,** another name for label correcting, just like Dijkstra's/A*.



RRT* Re-wiring

- After connecting vertex x_{new} to the tree, check alternative connections to all potential parents in ball radius r_n
- If a lower cost path found through different parent, add edge to new parent, delete edge to old parent
- If a lower cost path found through x_{new} as parent, add new/delete old
- r_n shrinks as n grows, according to

$$r_n = \gamma \left(\frac{\log n}{n} \right)^{\frac{1}{d}}$$



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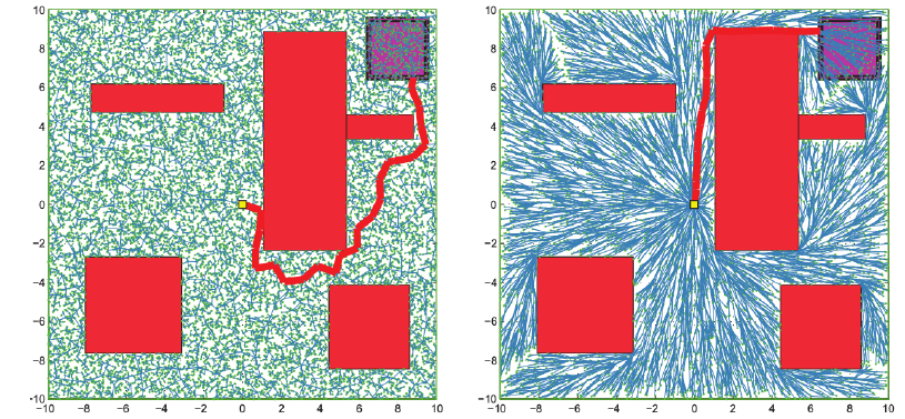

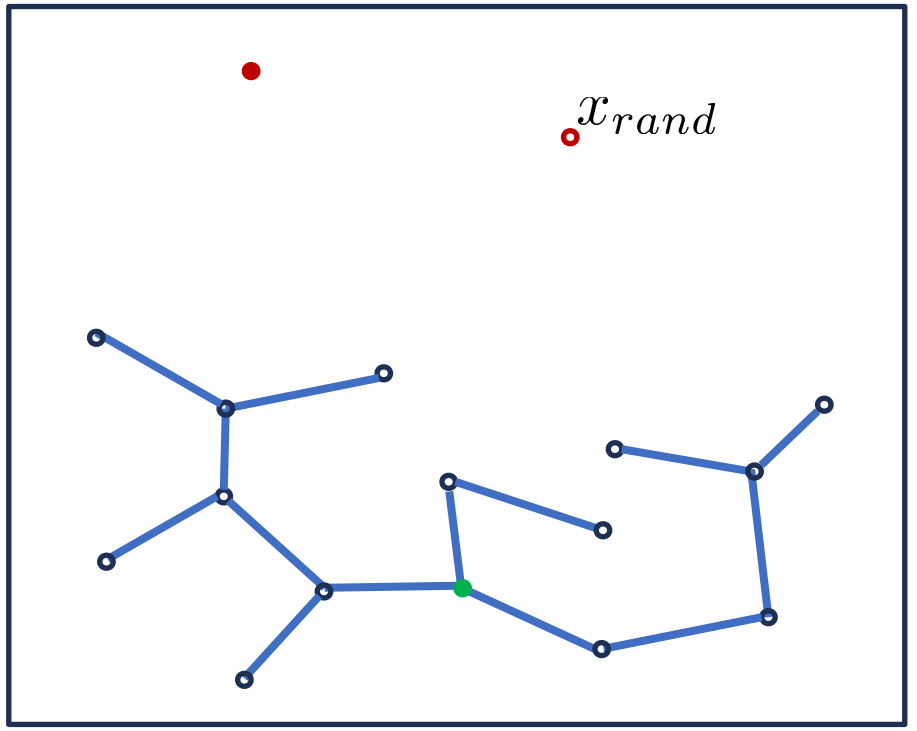

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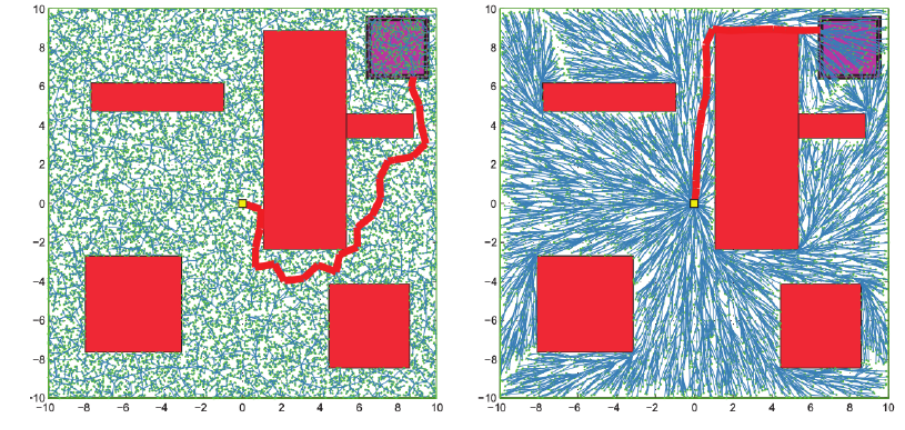

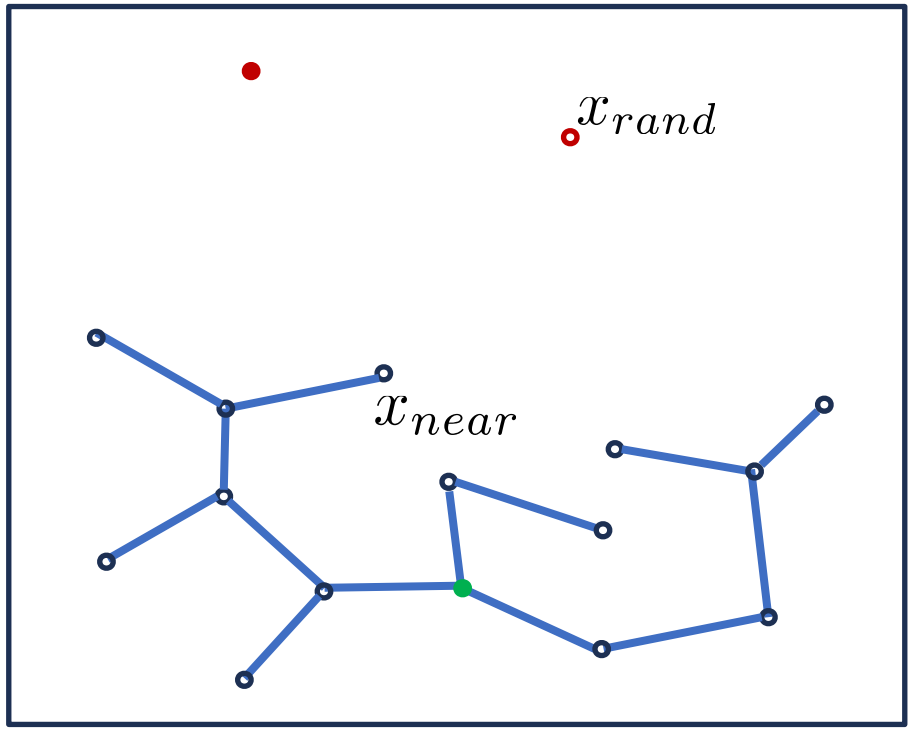

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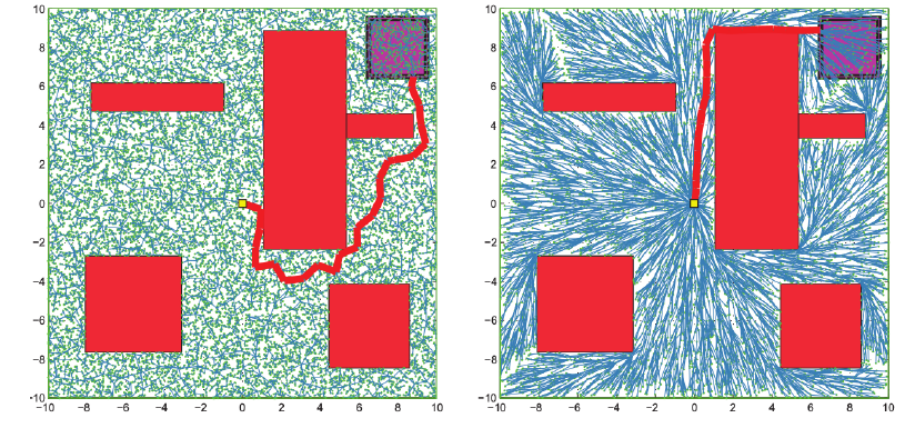

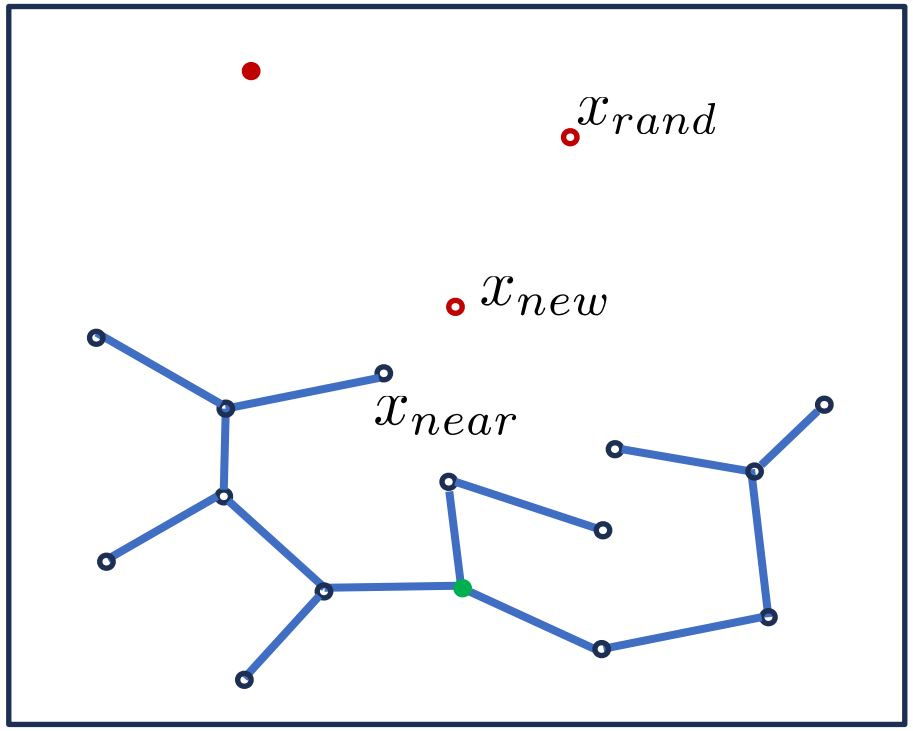

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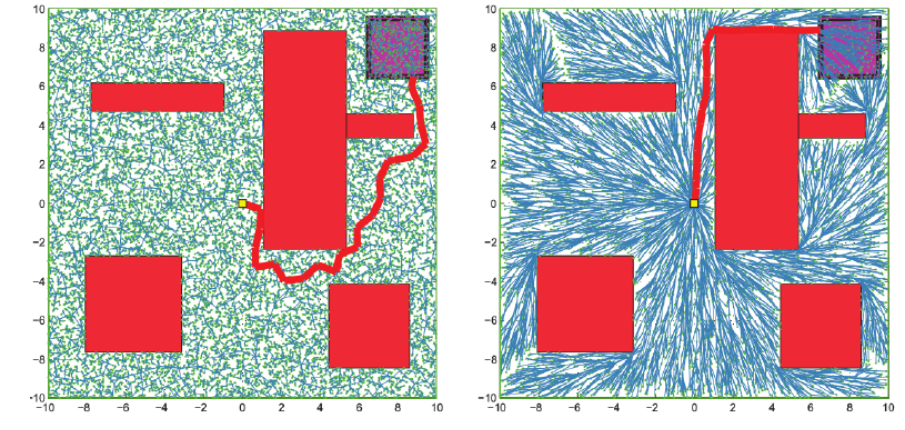

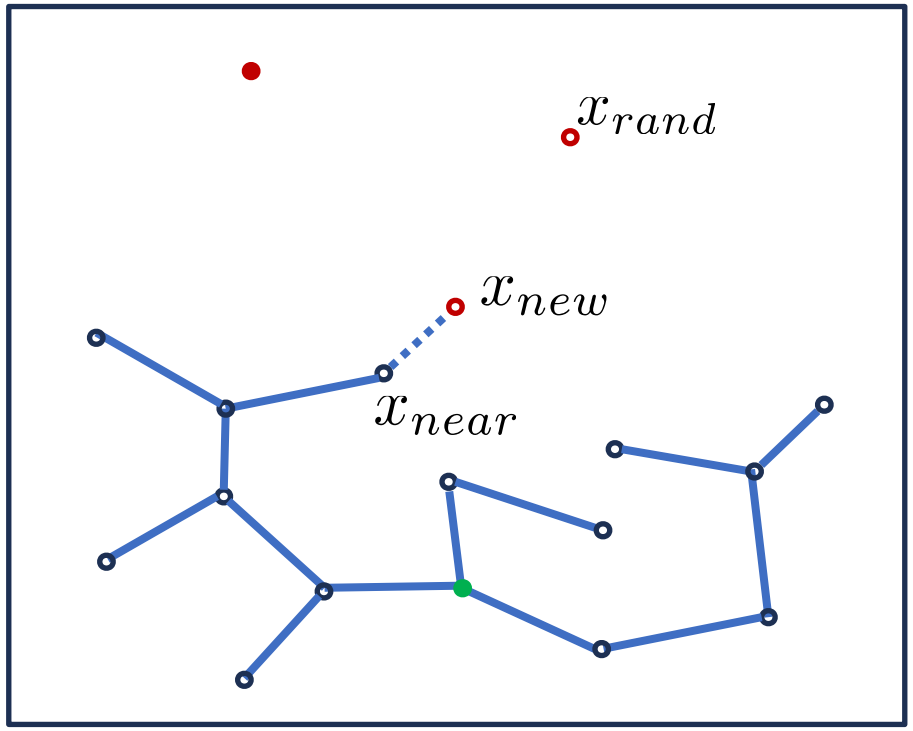

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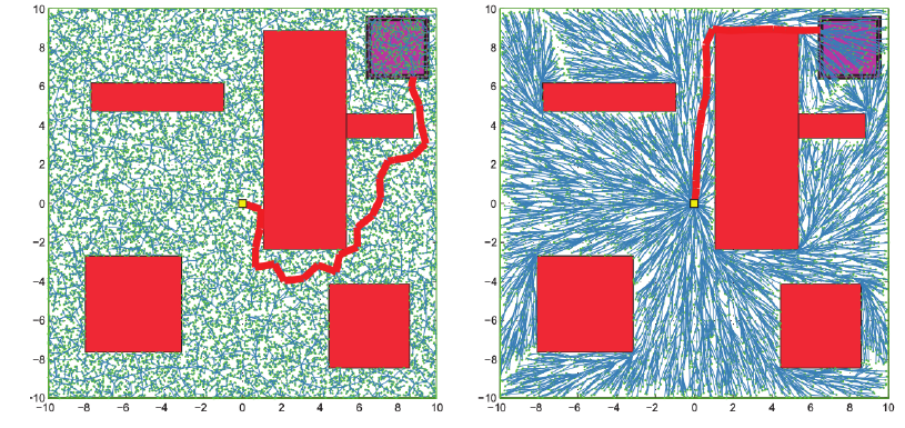

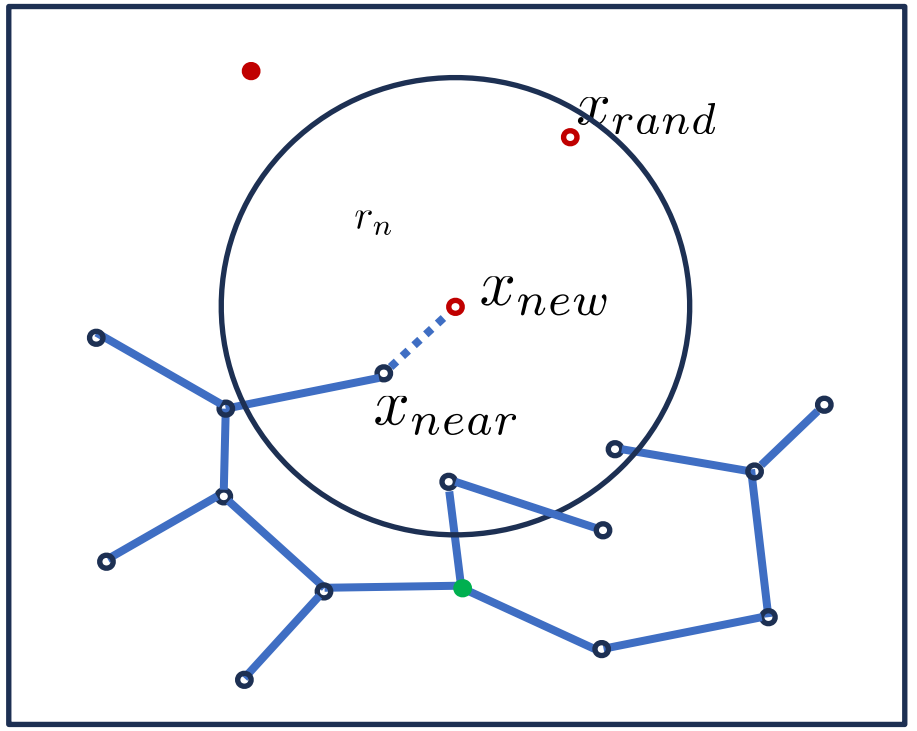

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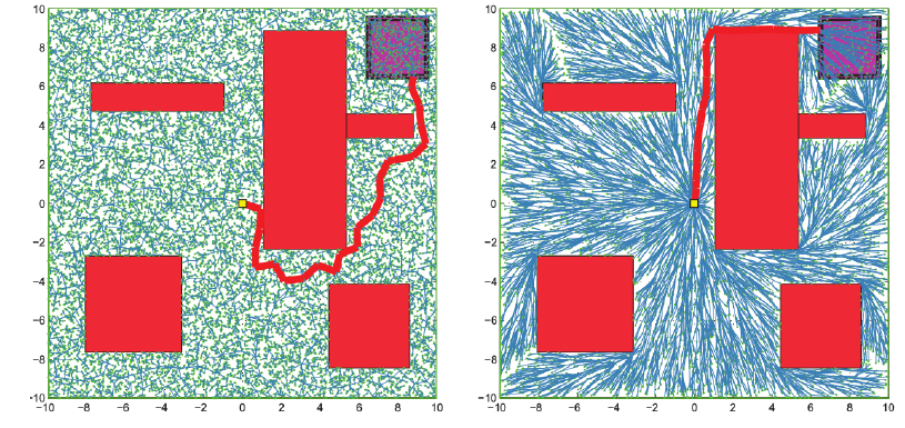

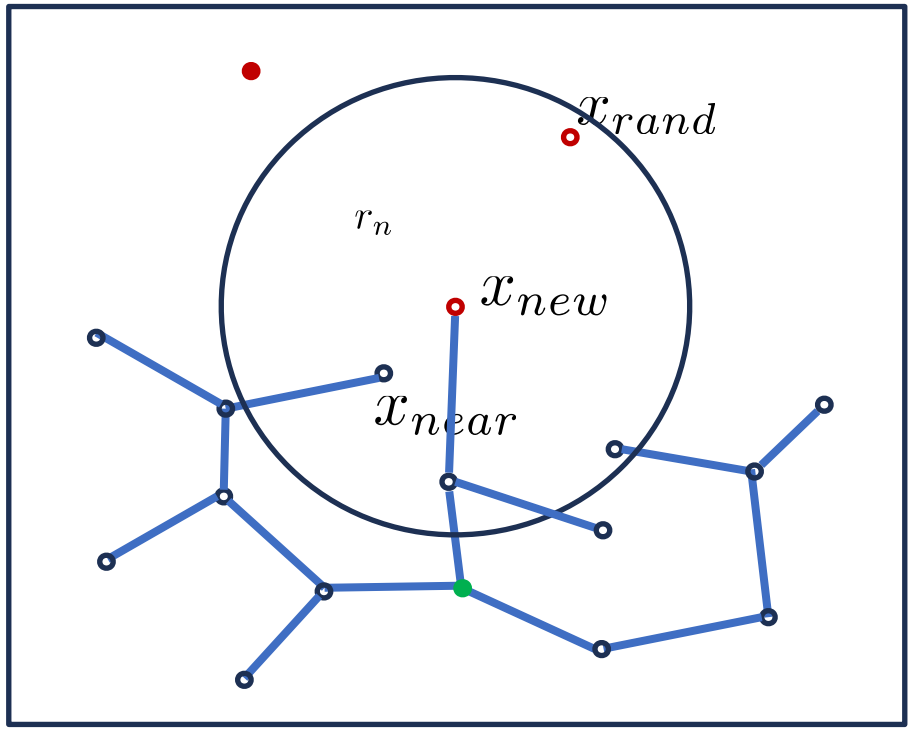

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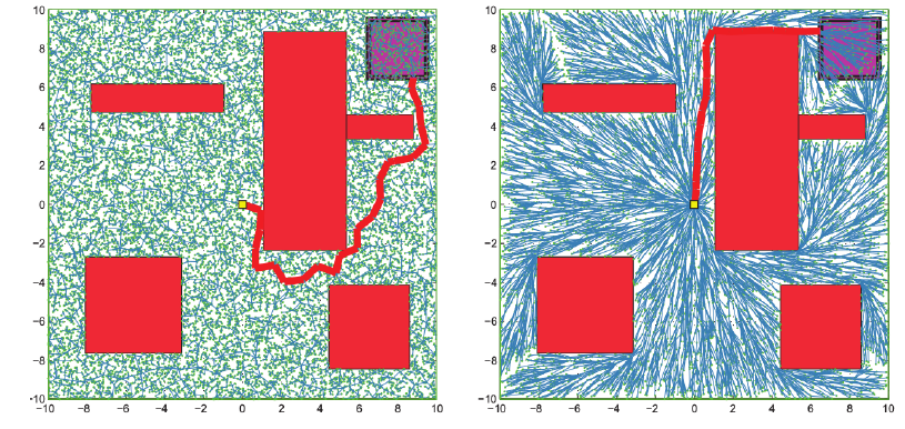
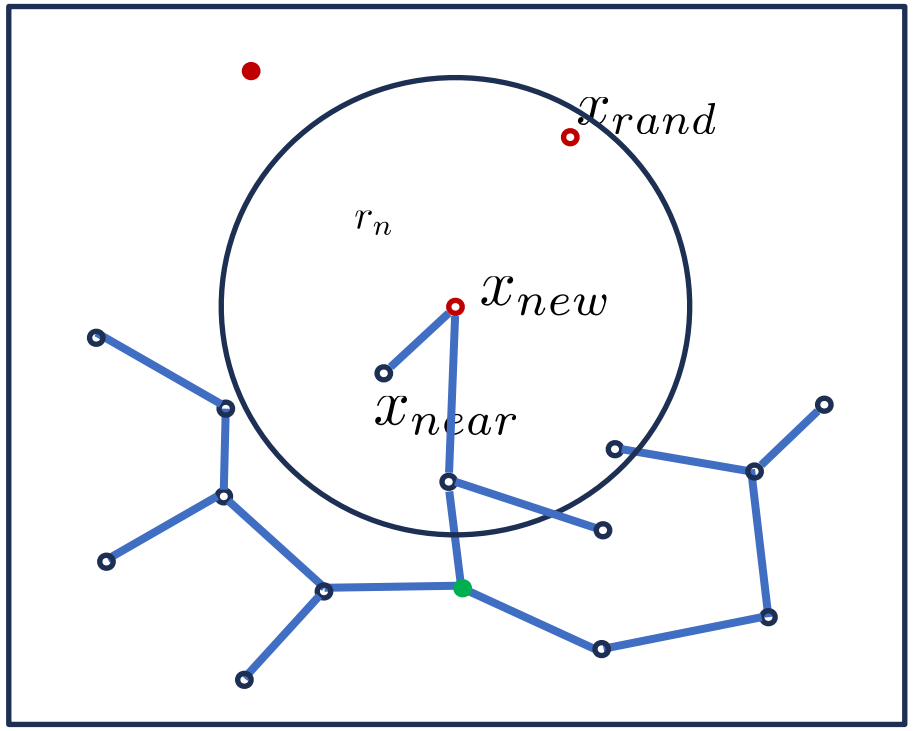




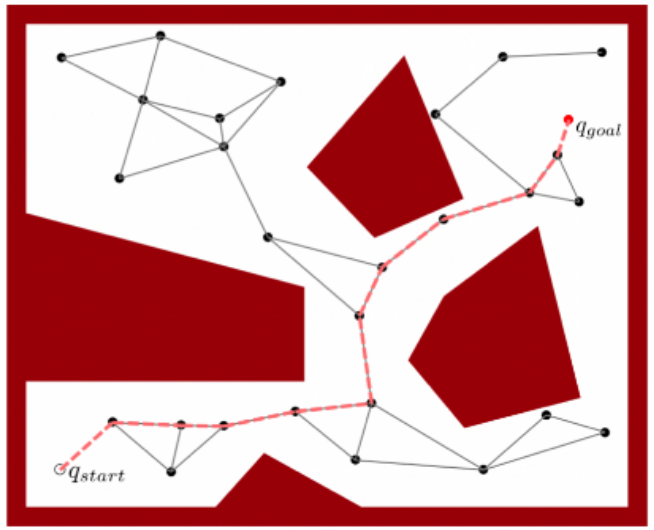
Observations

- PRM-like motion planning algorithms
  - For a given number of nodes n, they find “good” paths
  - ...however, require many costly collision checks

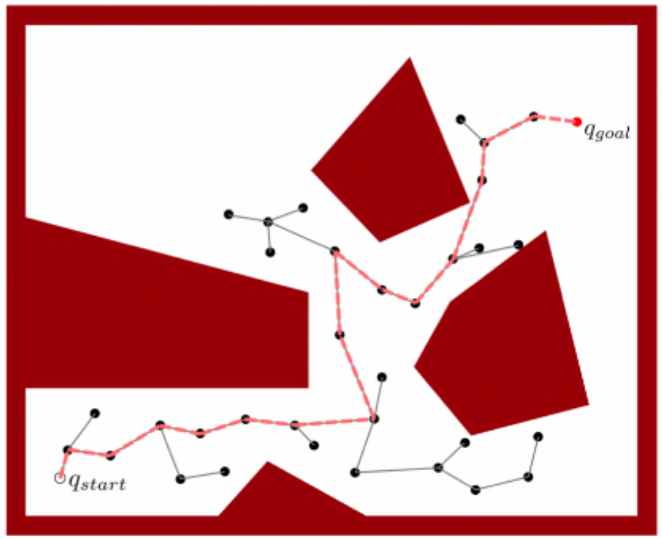
- RRT-like motion planning algorithms
  - Finds a feasible path quickly
  - ...however, the quality of that path is, in general, poor
  - “traps” itself by disallowing new better paths to emerge - RRT* performs local label correction as samples are added to help remedy this

Which of these two roadmaps has been created by RRT?

(A)



(B)



(A)

(B)

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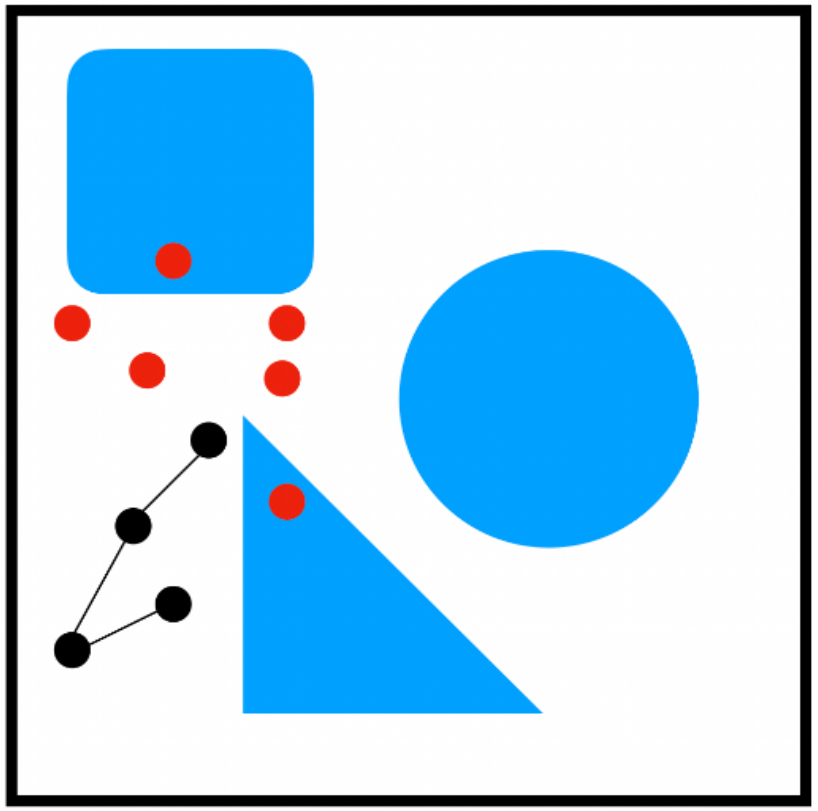
How many of the red sample points could lead to a new node in the roadmap constructed by RRT?



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Details and notes:

- Sampling-based planners transform the difficult global problem into a large set of local and easy problems
- A key ingredient is collision checking, which can be computationally time consuming. Speed ups avoid collision checking
- Local planning (edge validation) is typically performed by dense sampling of path and collision detection
- Another key ingredient is nearest-neighbor search (best done with k-d tree): given a query point find its nearest neighbor(s) within a set of points -- also well studied theoretically and practically

Sampling-based planning

Pros:

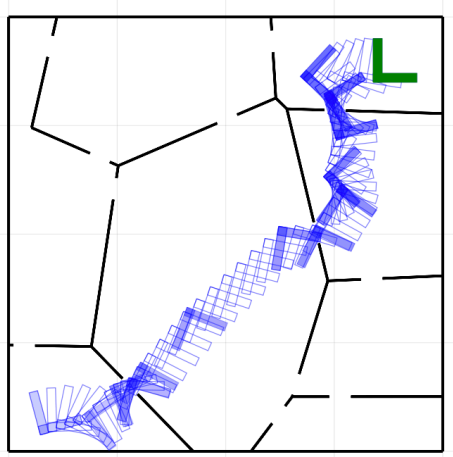
- Conceptually simple
- Relatively-easy to implement
- Flexible: one algorithm applies to a variety of robots and problems
- Beyond the geometric case: can cope with complex differential constraints, uncertainty, etc.

Cons:

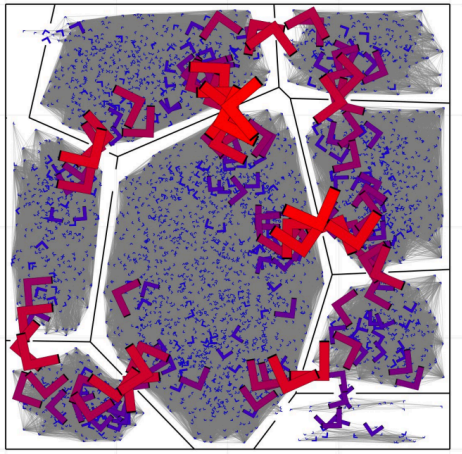
- Asymptotic
- Unclear how many samples should be generated to retrieve a solution
- Cannot determine whether a solution does not exist

Biased sampling for SBMP

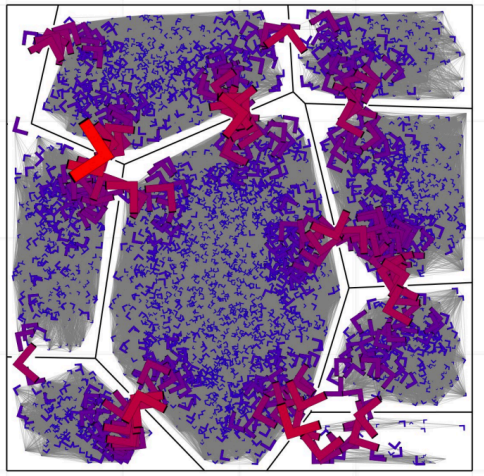
- Potential issue with uniform sampling: narrow corridors in C-space require many samples to identify/traverse
- Key idea: bias sampling towards suspected challenging regions of C-space
- Biased sampling distributions can be hand-constructed and/or adapt online (e.g., Hybrid Sampling PRM)
- Or learned from prior experience solving similar planning problems



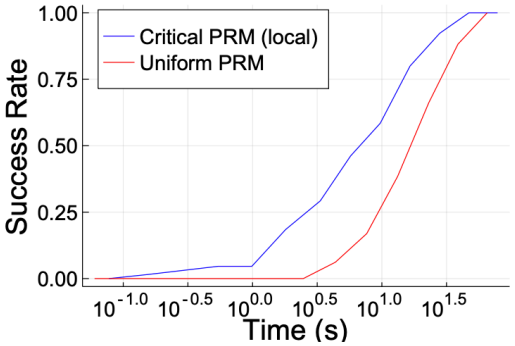
(a) SE(2) Planning



(b) Ground Truth Criticality



(c) Learned Criticality

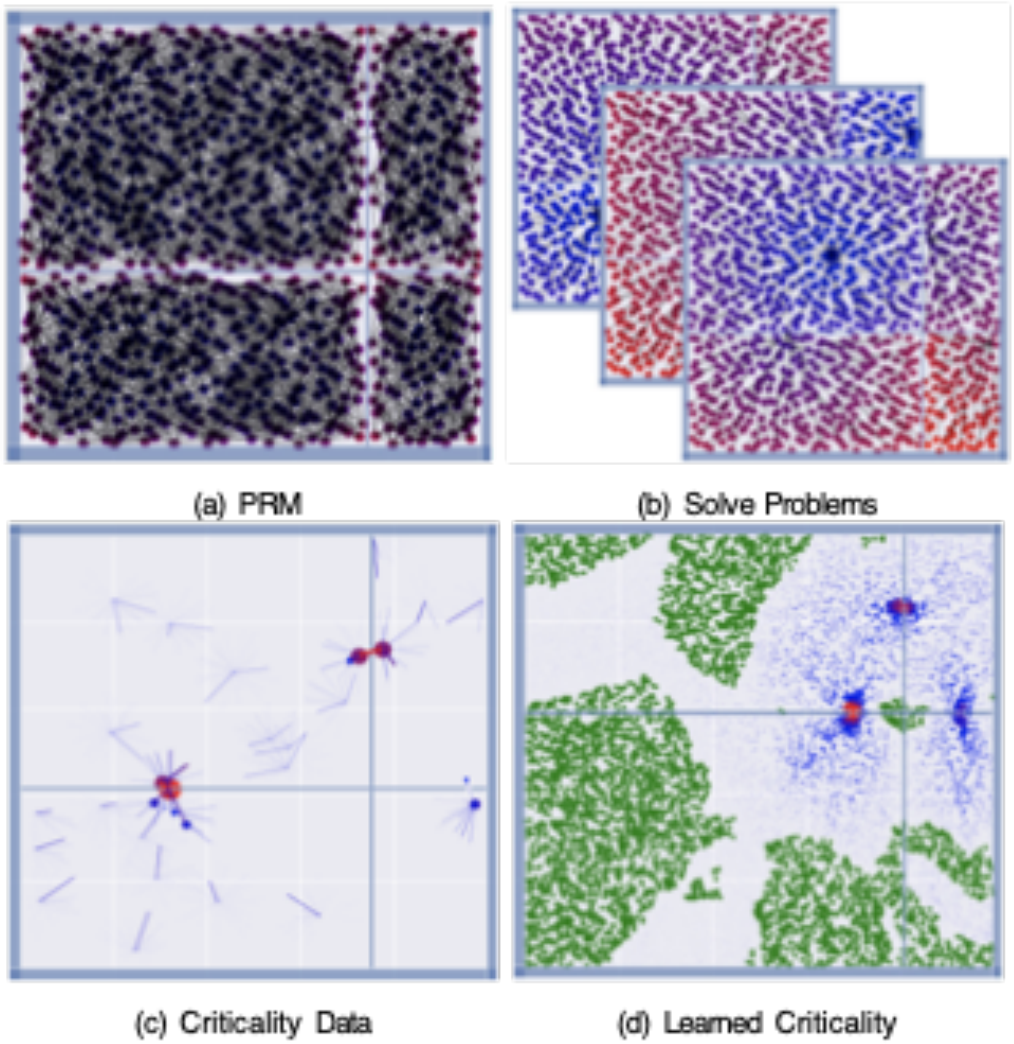


(d) Time (s) vs. Success

References:
Hsu et al. Hybrid PRM sampling with a cost-sensitive adaptive strategy. 2005.
Ichter et al. Learned Critical Probabilistic Roadmaps for Robotic Motion Planning. 2020

Learning criticality from local map features

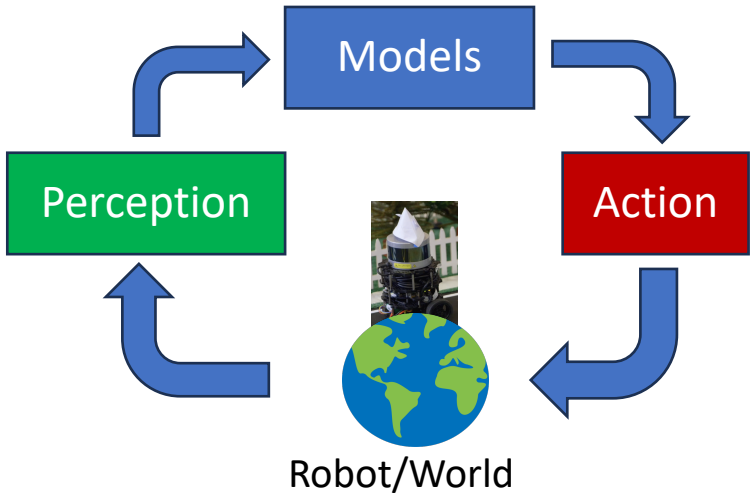
- On a collection of different maps, solve for optimal PRM paths between multiple start-goal pairs
- Score criticality of all nodes based on number of optimal paths passing through node
- Discount if node can be removed without collision
- Train CNN from local map patch centered at point to criticality of point



Ichter et al. Learned Critical Probabilistic Roadmaps for Robotic Motion Planning. 2020

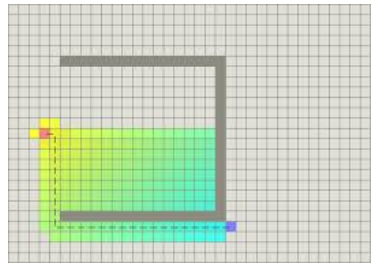
Action: Planning and Control Stack

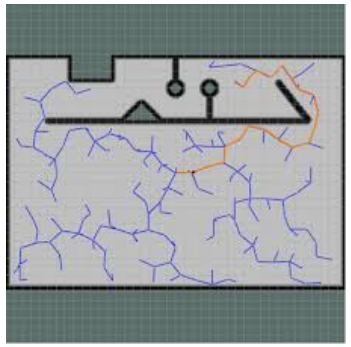
Use the models to move the robot



Path planning:

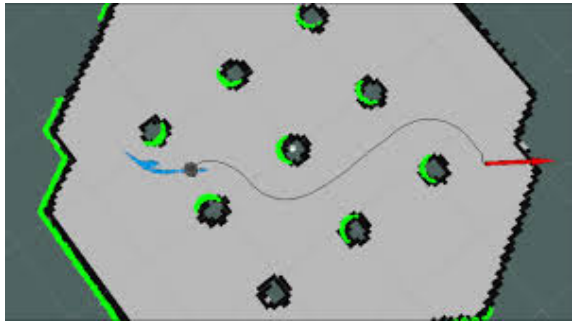
- A* planning on a graph
- RRT, RRT*





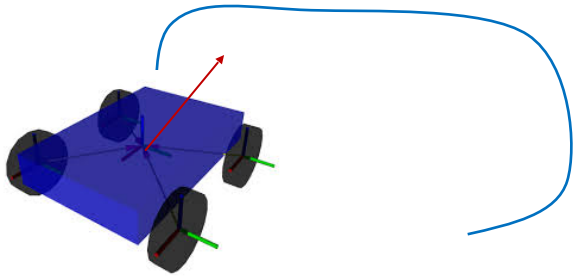
Trajectory optimization:

- Optimal control
- Numerical solvers



Trajectory following:

- Kinematic/dynamic models
- PID control
- LQR, gain scheduled LQR



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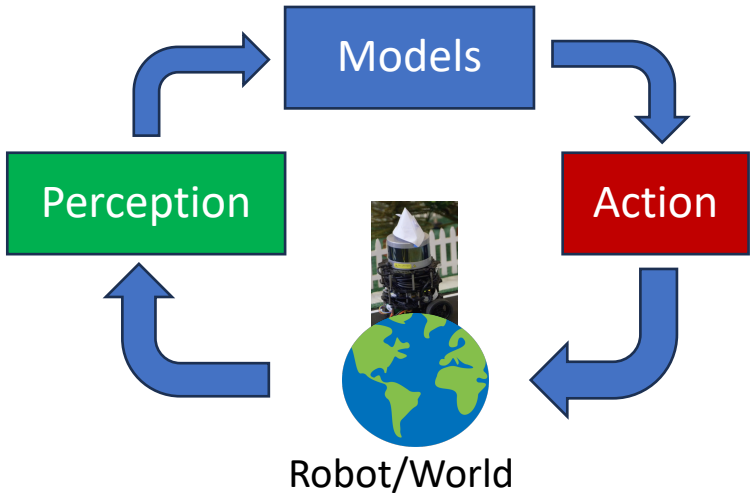

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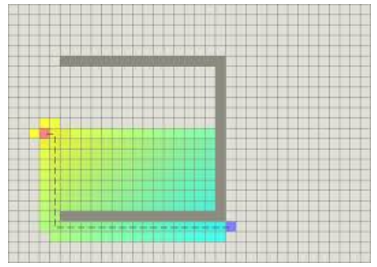

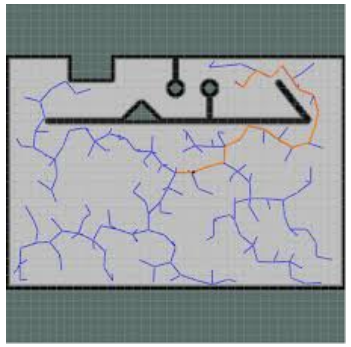

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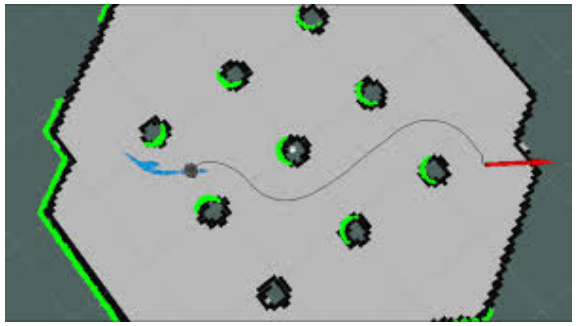

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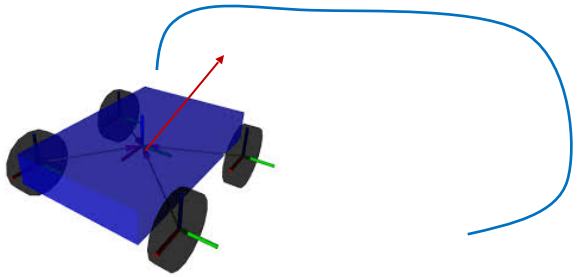
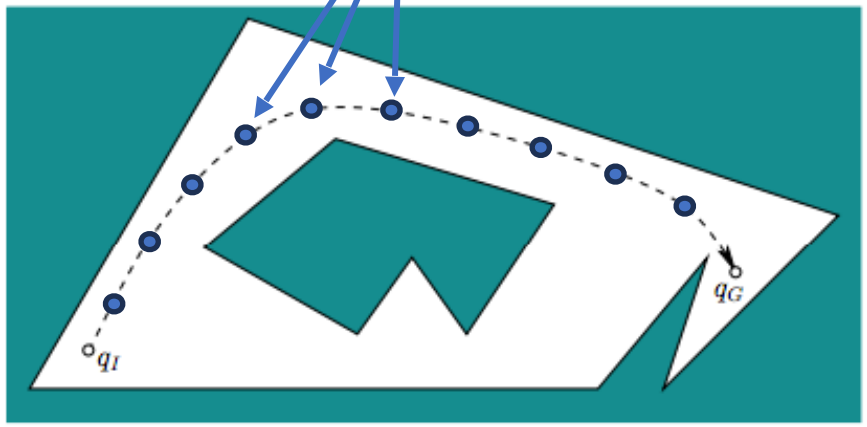


Trajectory Optimization

Waypoint poses from path planner
(e.g. A*, RRT*)



Trajectory optimization problem:

$$\min_{u_{0:T}} \sum_{t=0}^{T} c_t(x_t, u_t)$$

$$\text{s.t.} \quad x_{t+1} = f(x_t, u_t)$$

$$x_t \in \mathbb{X}_{\text{Free}}$$

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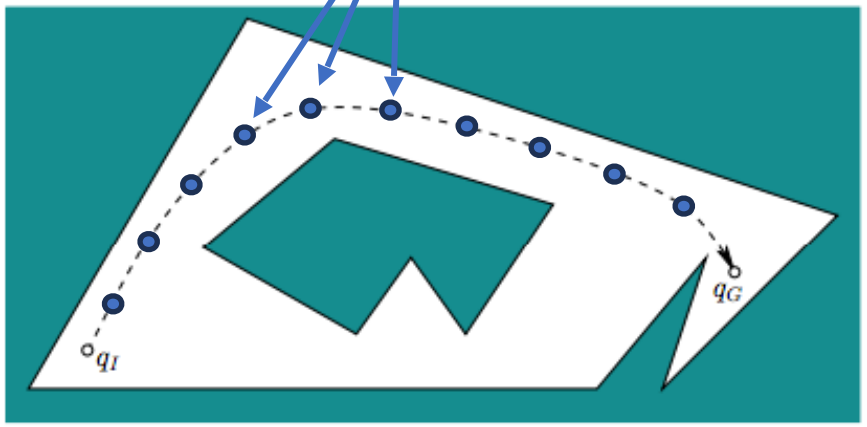

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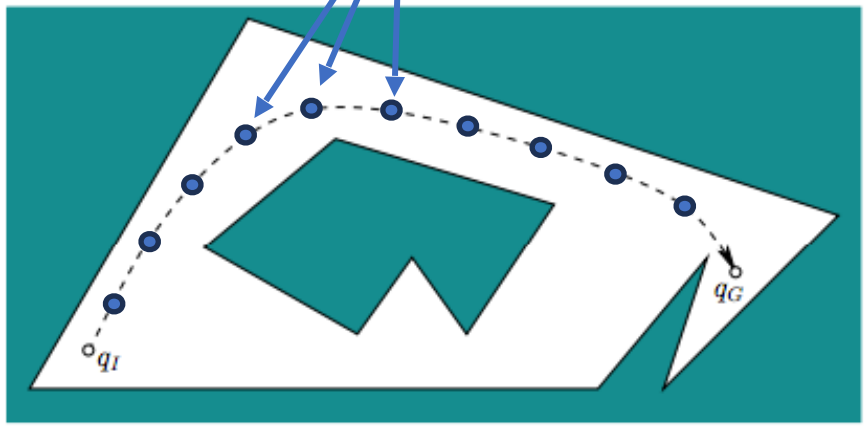

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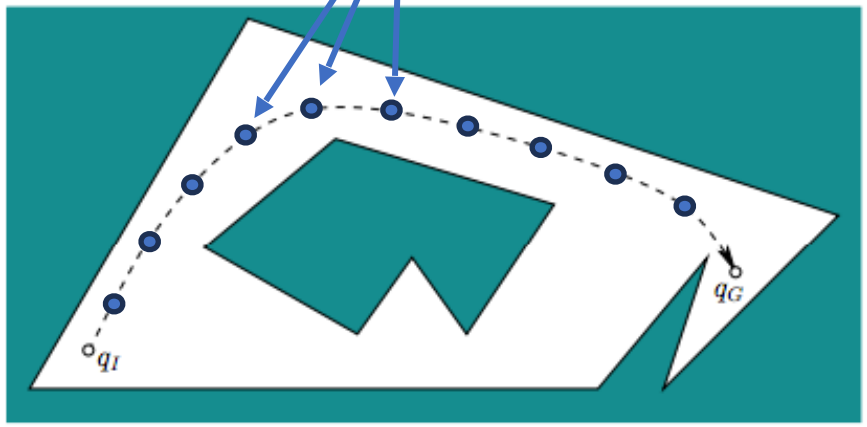

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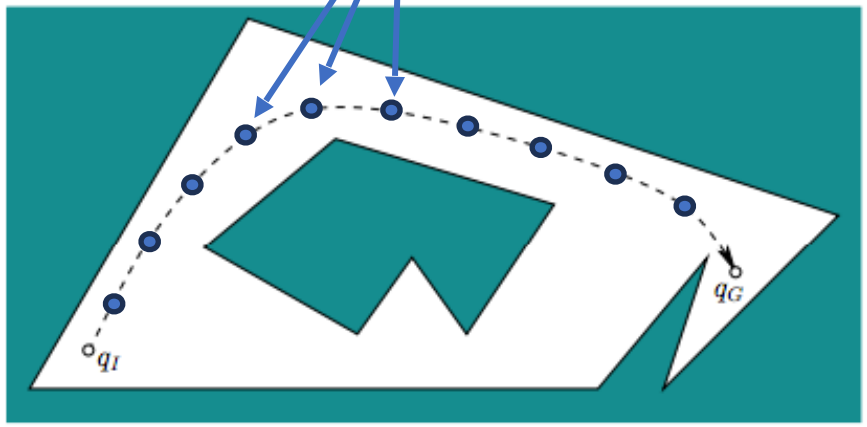

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$$x_t \in \mathbb{X}_{\text{Free}}$$ Collision constraints

Goal: Find a dynamically feasible trajectory (close to the path) so the robot doesn't hit things!

GuSTO: Guaranteed Sequential Trajectory Optimization via Sequential Convex Programming

Riccardo Bonalli, Abhishek Cauligi, Andrew Bylard, Marco Pavone

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Kinodynamic motion planning for vehicle high speed maneuvering

- Yanbo Li

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Admissible Velocity Propagation : Beyond Quasi-Static Path Planning for High-Dimensional Robots

Quang-Cuong Pham, Stéphane Caron, Puttichai Lertkultanon, Yoshihiko Nakamura. IJRR 2017.

Admissible Velocity Propagation : Beyond Quasi-Static Path Planning for High-Dimensional Robots

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Kinodynamic sampling based traj opt

- https://github.com/rikkimelissa/baxter_throw
- Rikki Valverde, Northwestern University

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Optimal control problem: traj opt and feedback control combined

The problem:

$$\begin{aligned} \min_{\mathbf{u}} \quad & h(\mathbf{x}(t_f), t_f) + \int_{t_0}^{t_f} g(\mathbf{x}(t), \mathbf{u}(t), t)\, dt \\ \text{subject to} \quad & \dot{\mathbf{x}}(t) = \mathbf{a}(\mathbf{x}(t), \mathbf{u}(t), t) \\ & \mathbf{x}(t) \in \mathcal{X}, \quad \mathbf{u}(t) \in \mathcal{U} \end{aligned}$$

where $\boldsymbol{x}(t) \in R^n$, $\boldsymbol{u}(t) \in R^m$, and $\boldsymbol{x}\left(t_0\right) = \boldsymbol{x}_0$

- Good reference: D. K. Kirk. Optimal Control Theory: An introduction. 2004.

Form of optimal control

- **Closed loop:** If a functional relationship can be found in form

$$\mathbf{u}^*(t) = \pi(\mathbf{x}(t), t)$$

- **Open loop:** If the optimal control law is determined as a function of time for a specified initial state value

$$\mathbf{u}^*(t) = \mathbf{f}(\mathbf{x}(t_0), t)$$

- A good compromise: two-step design

$$\mathbf{u}^*(t) = \mathbf{u}_d(t) + \pi(\mathbf{x}(t), \mathbf{x}(t) - \mathbf{x}_d(t))$$

Reference trajectory

Reference control (open-loop), from traj opt

Trajectory-tracking law (closed-loop), from feedback control

Tracking error

Open-loop control

- We want to find

$$\mathbf{u}^*(t) = \mathbf{f}(\mathbf{x}(t_0), t)$$

- In general, two broad classes of methods:

  1. Direct methods: transcribe infinite problem into finite dimensional, nonlinear programming (NLP) problem, and solve NLP $\Rightarrow$ "First discretize, then optimize"
  2. Indirect methods: attempt to find a minimum point "indirectly," by solving the necessary conditions of optimality $\Rightarrow$ "First optimize, then discretize"

- For an in-depth study of direct and indirect methods, see AA203 "Optimal and Learning-based Control"

Extra slides

Should probabilistic planners be probabilistic?

Key question: would theoretical guarantees and practical performance still hold if these algorithms were to be de-randomized, i.e., run on deterministic samples?

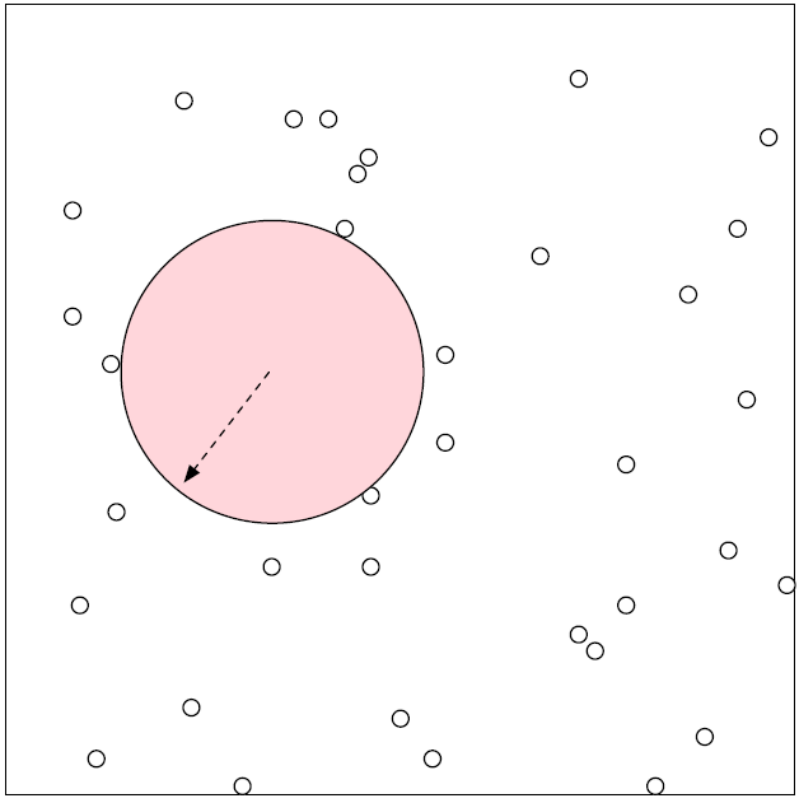
Important question as de-randomization would:

- Ease certification process
- Ease use of offline computation
- Potentially simplify several operations (e.g., NN search)

Designing "good" sequences

ℓ_2-dispersion: For a finite set S of points contained in $X \subset \mathbb{R}^d$, its ℓ_2-dispersion $D(S)$ is defined as

$$D(S) := \sup_{x \in X} \min_{s \in S} \|s - x\|_2$$



Key facts:

- There exist deterministic sequences with $D(S)$ of order $O\left(n^{-1/d}\right)$, referred to as **low-dispersion** sequences
- Sequences minimizing ℓ_2-dispersion only known for $d = 2$

Optimality of deterministic planning

```
1  V ← {x_init} ∪ SampleFree(n); E ← ∅
2  for all v ∈ V do
3     X_near ← Near(V\{v}, v, r_n)
4     for x ∈ X_near do
5        if CollisionFree(v, x) then
6           E ← E ∪ {(v, x)} ∪ {(x, v)}
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10 return ShortestPath(x_init, V, E)
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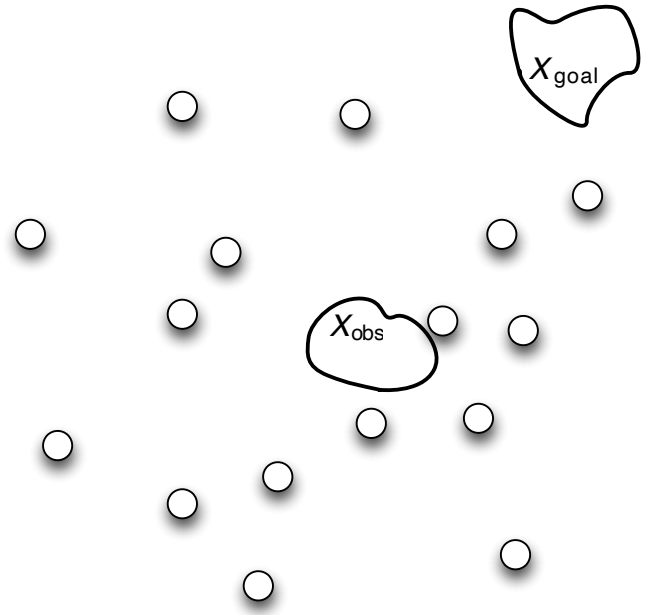

Optimality: Let c_n denote the arc length of the path returned with n samples. Then if

1. Samples set S has dispersion $D(S) \leq \gamma n^{-1/d}$ for some $\gamma > 0$,
2. $n^{1/d} r_n \to \infty$,

then $\lim_{n \to \infty} c_n = c^*$, where c^* is the cost of an optimal path

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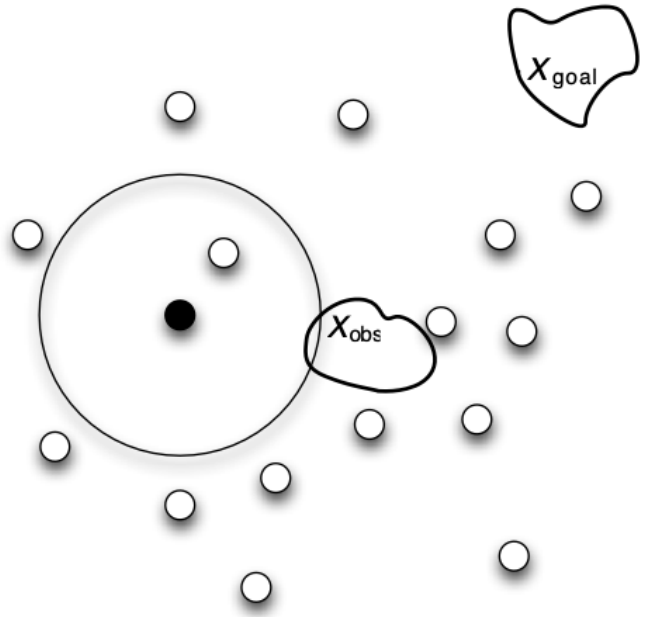

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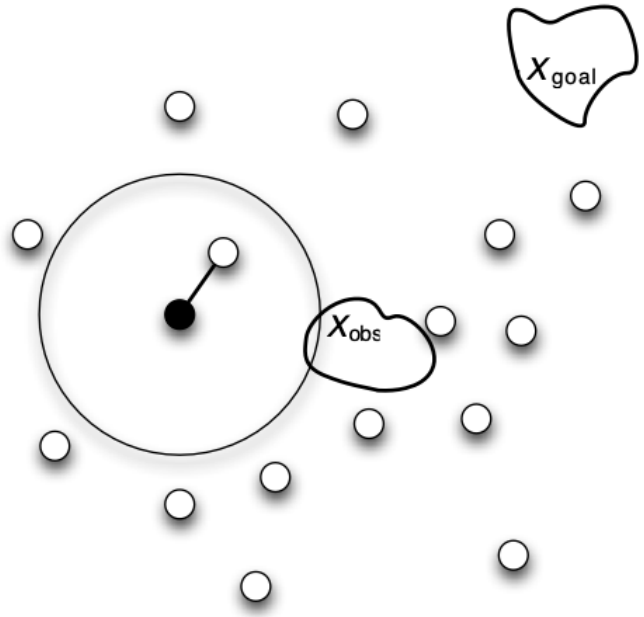

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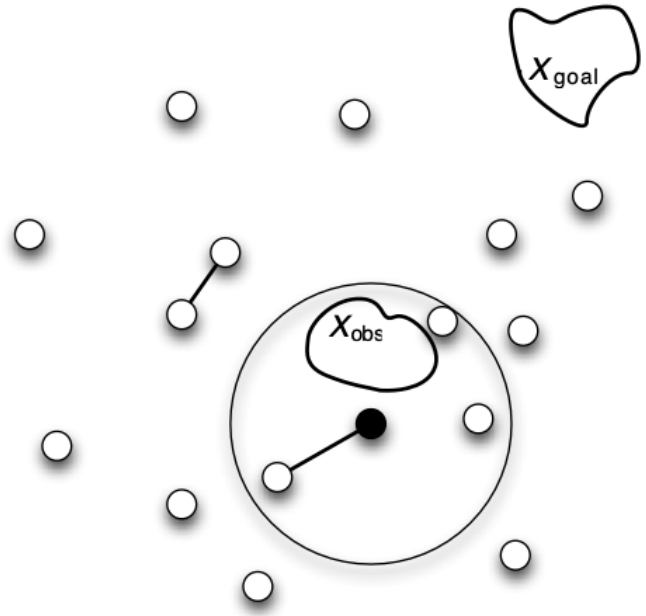

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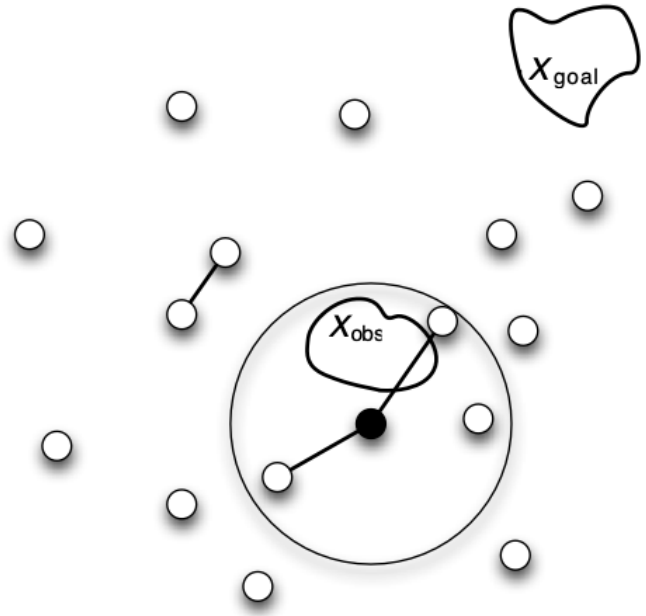

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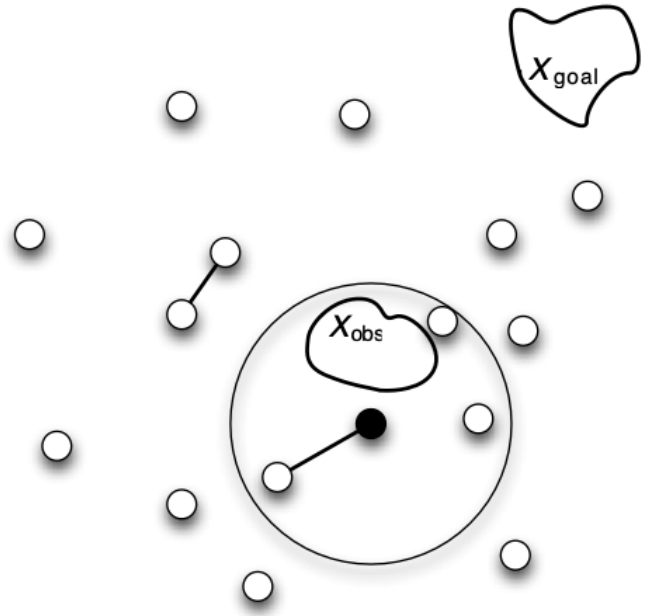

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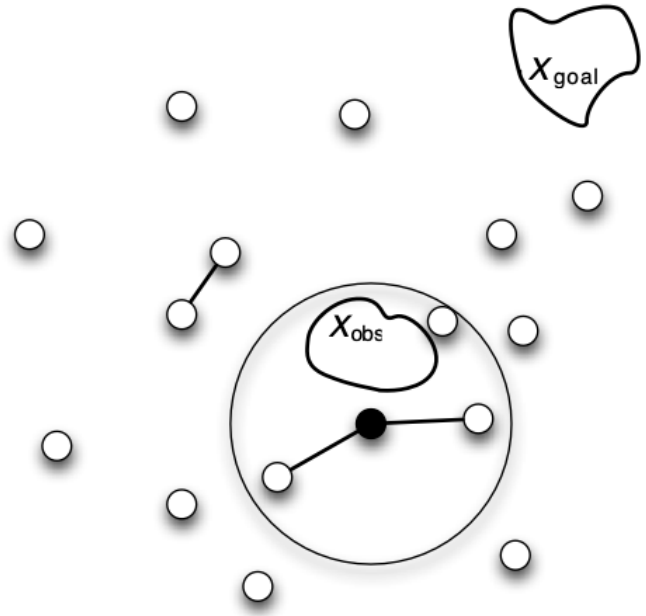

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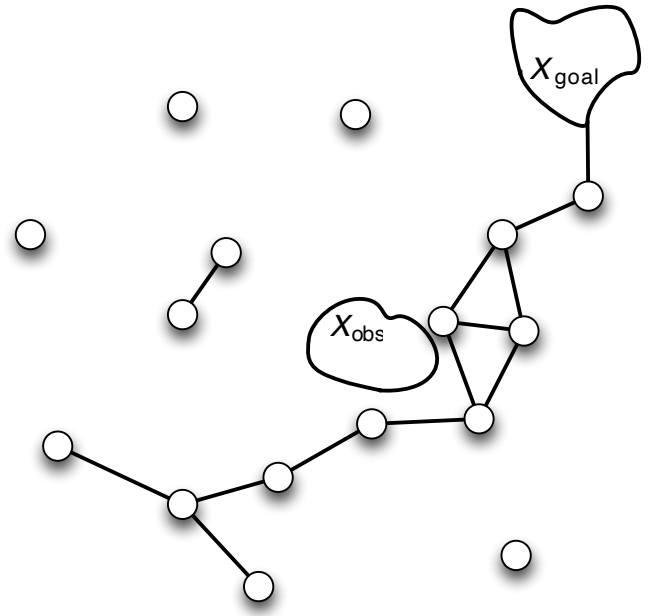

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1. Samples set S has dispersion $D(S) \leq \gamma n^{-1/d}$ for some $\gamma > 0$,
2. $n^{1/d} r_n \to \infty$,

then $\lim_{n \to \infty} c_n = c^*$, where c^* is the cost of an optimal path

Deterministic sampling-based motion planning

- Asymptotic optimality can be achieved with deterministic sequences and with a smaller connection radius
- Deterministic convergence rates: instrumental to the certification of sampling-based planners
- Computational and space complexity: under some assumptions, arbitrarily close to theoretical lower bound
- Deterministic sequences appear to provide superior performance

Reference: Janson et al. Deterministic Sampling-Based Motion Planning: Optimality, Complexity, and Performance. 2018